Beginnings in Algebraic Geometry Notes for CH 9 - 12

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1 CH 9 Projective Varieties

1.1 9.1 Projective Space

Definition 1.1 (Projective Space)**.** *Let $n \in \mathbb{N}$ a projective space over K denoted $\mathbb{P}^n_k$ is defined as*

$$\frac{K^{n+1}/\{(0,\dots,0)\}}{\sim}$$

where $(a_0,\dots,a_n) \sim (b_1,\dots,b_n)$ iff there exists a $\lambda \in K/\{0\}$ such that

$$(a_0,\dots,a_n) = (\lambda b_0,\dots,\lambda b_n).$$

An equivalence class is denoted $[a_0 : \cdots : a_n]$.

Remark 1.1. Note here that we aren't really putting any structural requirements on K. My previous encounters with projective spaces almost always used a field and the notation of K is consistent with the textbooks use of fields. With affine spaces we noted that the Nullstellensatz was only consistent when K was an algebraically closed field. I imagine we will soon face a similar restriction. *Does K need to be a field?*

Projective spaces are particularly interesting because there are many related ways of presenting them. The first thing we'll do as a responsible mathematicitians is go through a couple of examples.

Ex Let $n = 0$, that is $\mathbb{P}^0_K$ for some K. This means we are looking at equivalence classes of $\frac{K/\{0\}}{\sim}$. Note that for any $a \in K/\{0\}\}$ we can always take $\lambda = \frac{1}{a}$ therefore all of $\mathbb{P}^0 = \{[1]\}$.

Ex Take $n = 1$, for any $[a_0 : a_1] \in \mathbb{P}^1$ observe that by taking $\lambda = \frac{1}{a_0}$ then every equivalence class where $a_0 \neq 0$ is simply $[1 : \frac{a_1}{a_0}]$. On the other hand when the first term is zero, $[0 : a_1]$ we may always choose $\lambda = \frac{1}{a_1}$ so that it may be scaled to $[0 : 1]$. This means that $\mathbb{P}^1$ can instead be expressed as a disjoint union

$$\mathbb{P}^1 = \{[1 : a] \mid a \in K\} \sqcup \{[0 : 1]\}.$$

Observe that the left term is in bijection with $\mathbb{A}^1$

Ex Let $n = 2$. Similar to $\mathbb{P}^1$, equivalence classes $[a_0 : a_1 : a_2] \in \mathbb{P}^2$ can always be rescaled to $[1 : b_1 : b_2]$ when $a_0 \neq 0$. If the first term is zero then again every term can be scaled to $[0 : b_1 : b_2]$. Then,

$$\mathbb{P}^2 = \{[1 : b_1 : b_2] \mid b_1, b_2 \in K\} \sqcup \{[0 : b_1 : b_2] \mid b_1, b_2 \in K\}$$

Interestingly the by dropping the first term; the left portion is in bijection with $\mathbb{A}^2$ while the right term is in bijection with $\mathbb{P}^1$ which gives the decomposition

$$\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{P}^1.$$

Theorem 1.2. *For $n \geq 1$ that there exists a natural bijection $\mathbb{P}^n = \mathbb{A}^n \sqcup \mathbb{P}^{n-1}$*

Proof. The proof follows the structure provided in problem 9.1.3 and consists of three portions. First define

$$U = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_0,\dots,a_n \in K, a_0 \neq 0\} \quad \text{and} \quad V = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_0,\dots,a_n \in K, a_0 = 0\}$$

We begin by showing that $\mathbb{P}^n = U \sqcup V$. The containment of $\mathbb{P}^n$ in $U \cup V$ and vice verse is automtic given the definition of U and V. It remains to show that $U \cup V$ can be upgraded to $U \sqcup V$. Suppose towards a contradiction that $U \cap V \neq \emptyset$ then there exists $[b_0 : \cdots : b_n] \in U \cap V$. Note that for any $[a_1 : \cdots : a_n] \in U$, because a_0 is non-zero, we may always take $\lambda = \frac{1}{a_0}$ so that every equivalence class contained in U is of the form $[1 : c_1 : \cdots : c_n]$ for some $c_1,\dots,c_n \in K$. The assumption that $U \cap V$ is non-empty implies that b_0 is both zero and one. Hence $U \cap V = \emptyset$ and $\mathbb{P}^n = U \sqcup V$. The isomorphism $V \simeq \mathbb{P}^{n-1}$ and $U \simeq \mathbb{A}^n$ follow immediately by simply dropping the leading zero or one. ℵ

There are a few more ways that projective spaces are represnted. The most well known is that $\mathbb{P}^n$ is isomorphic to the set of points through the origin of $\mathbb{A}^{n+1}$

1.2 9.2 Projective Vanishing Operator

We say that a polynomial $f \in K[x_0, \ldots, x_n]$ only vanishes on a point a if it vanishes on every representative of a in the desired projective space.

Non-Example: Consider the polynomnial $f(x, y) = x^2 - y$ which vanishes on $[2, 4]$ but it doesnt vanish on $[4, 8]$

Example: The polynomial $f(x, y) = x^2 - y^2$ on the other hand vanishes on $[2, 2]$ and every possible representative of the solution

We end up with a problem: How do we check that every representative vanishes? Instead of checking we simply restrict ourselves to the case of homogenous polynomials. A polyomial is considered homogenous of degree d if every non-zero term of f has degree d. For example $f(x, y, z) = 3x^2 + 2yz + 3zx$ is a polynomial homogenous of degree 2 because every non-zero term has degree 2.

Proposition 1.1. *Let $f \in K[x_0, \ldots, x_n]$ be a homogenous polynomial of degree d and $[a_0 : \cdots : a_n] = [b_0 : \cdots : b_n]$ in $\mathbb{P}^n$ then $f(a_0, \ldots, a_n) = 0 \iff f(b_0, \ldots, b_n) = 0$.*

Proof. If f is a homogenous polynomial of degree d, then each non-zero term of f takes the form $cx_0^{\alpha_0} \cdots x_n^{\alpha_n}$ for some non-zero $c \in K$, $\alpha_0, \ldots, \alpha_n \in \mathbb{N}$ such that $\sigma\alpha_i = d$. Evaluating f at a representative of $[a_0 : \cdots : a_n]$ means each non-zero term takes the form

$$c\lambda^{\sum_i \alpha_i} a_0 \cdots a_n = c\lambda^d a_0 \cdots a_n.$$

Then λ^d can be factored out of every non-zero term of f evaluating at so that $f(\lambda a_0, \ldots, \lambda a_n) = \lambda^d f(a_0, \ldots, a_n)$. If $f(a_0, \ldots, a_n) = 0$ and, by assumption, there exists a $\lambda \in K$ such that

$$(\lambda a_0, \ldots, \lambda a_n) = (b_1, \ldots, b_n)$$

with the desired implication following direcrtly. Following identical logic the reverse implication completes the desired if and only if. ℵ

Of course in true algebraist sense, the fact that we can determine that a homegenous polynomial vanishes on all representatives as long as it vanishes on one, we restrict ourselves to looking only at homogenous polynomials. Yet we would like to still work with non-homogenous as a strict homogeneity condition is too restrictive. Lucky for us we can considere homogenous components of a polynomial. For example the polynomial $f(x, y) = x^2 + xy + x$ has a homogenous component of degree 2 and degree 1 denoted

$$f_2(x, y) = x^2 + xy \quad \text{and} \quad f_1(x, y) = x$$

. The next proposition follows quite naturally

Proposition 1.2. *Let $f \in K[x_0, \ldots, x_n]$ be a polynomial and $a \in \mathbb{P}^n$. Then f vanishes at a if and only if each homogenous component of f vanishes at a.*

Proof. Suppose that each homogenous component f_k of f vanishes on a for all $k \leq d$ where d is the degree of polynomial f. As

$$f = \sum_{k=1}^{d} f_k$$

and $f_k(a) = 0$ for every representative of a, then f vanishes at every representative of a. Now suppose that f vanishes on every representative of a. Then for any $\lambda \in K \setminus \{0\}$, by assumption it follows

$$0 = f(\lambda a_0, \ldots, \lambda a_n) \tag{1}$$

$$= \sum_{k=1}^{d} \lambda^d f_k(a_0, \ldots, a_n). \tag{2}$$

Since (2) vanishes of inifnitely many λ we may view (2) as a univariate polynomial

$$f(x) = \sum_{k=1}^{d} x^d f(a_0, \ldots, a_n) = 0$$

vanishes on infinitely many values of K. This implies that $f(x)$ must be the zero polynomial then, $f_k(a_0, \ldots, a_n) = 0$ for all k. Since f_k is homogenous then each homogenous component of f must vanish on every representative of a. ℵ

Definition 1.2 (Projective $\mathcal{V}$ Operator)**.** *Let $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$ be a set of polynomials. The projective vanishes set of $\mathcal{S}$ is*

$$\mathcal{V}_{\mathbb{P}}(\mathcal{S}) = \{a \in \mathbb{P}^n \mid f(a) = 0 \; \forall f \in \mathcal{S}\}$$

We say that $X \subseteq \mathbb{P}^n$ is projective variety if $X = \mathcal{V}_{\mathbb{P}}(S)$ for some set $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$.

Analagously to the affine case, a projective variety is defined by a ideals, i.e.

Proposition 1.3. *If $S \subseteq K[x_0, \ldots, x_n]$ is a set of polynomials then*

$$\mathcal{V}_{\mathbb{P}}(\mathcal{S}) = \mathcal{V}_{\mathbb{P}}(\langle \mathcal{S} \rangle)$$

Proof. Let $a \in \mathcal{V}_{\mathbb{P}}(\mathcal{S})$, implies that for any polynomial $f \in \mathcal{S}$ that f vanishes on a. Let $g \in K[x_0, \ldots, x_n]$ be a polynomial. The product fg evaluated at a must also vanish. Therefore $fg \in \mathcal{S}$ hence $\mathcal{S}$ must be an ideal. The reverse inclusion comes immediately. ℵ

Further specifying, again like the affine case, we see that projective varieties are finitely generated by homogenous polynomials.

Proposition 1.4. *Any projective variety $X \subseteq \mathbb{P}^n$ is of the form $X = \mathcal{V}_{\mathbb{P}}(f_1, \ldots, f_k)$ where $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ are homogenous polynomials.*

Proof. Let $X \subseteq \mathbb{P}^n$ be a projective variety then $X = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$ where $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$ is an ideal. Hilbert's Basis theorem ensures that $K[x_0, \ldots, x_n]$ is Neotherian implying there exist polynomials $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ such that $\mathcal{S} = \langle f_1, \ldots, f_k \rangle$. Using lemma 9.16 note that if each polynomial f_i of degree d vanishes on a point $a \in \mathbb{P}^n$ then every homogenous component $f_{i,k}$ of degree k for $k \leq d$ must also vanishes on a. Since $f_i = \sum_p = 1^d f_{i,p}$ then we may further refine our list of generating polynomials. Hence any projective variety is the vanishing of a finite set of homogenous polynomials. ℵ

1.3 9.3 Projective $\mathcal{I}$-operator

It is defined essentially identically to the affine vanishing operator

Definition 1.3 (Projective $\mathcal{I}$-operator)**.** *Let $X \subseteq \mathbb{P}^n$ be a subset. The vanishing ideal of X is the set*

$$\mathcal{I}_{\mathbb{P}}(X) = \{f \in K[x_0, \ldots, x_n] \mid f(a) = 0 \; \forall a \in X\}$$

Ex: Let $X = \{[1:0]\} \subseteq \mathbb{P}^n$. Observe that for the polynomial $f(x_0, x_1) = x_1$ any representative of $[1:0]$ will always vanish. In fact we will show that $\langle x_1 \rangle = \mathcal{I}_{\mathbb{P}}(X)$. The inclusion $\langle x_1 \rangle \subseteq \mathcal{I}_{\mathbb{P}}(X)$ follows immediately as x_1 is fixed to zero. Now take $f \in \mathcal{I}_{\mathbb{P}}(X)$ then $f(a, 0) = 0$ for all $a \in K$, then we can view the polynomial as univariate in the first coordinate $f(x_0, 0)$ with infinitely many zeroes implying it must be the zero polynomial. Take a polynomial $f \in K[x_0, x_1]$ and view it as a univariate polynomial $f \in (K[x_0])[x_1]$ which has a general form

$$f = \sum_{d \geq 0} f_d(x_0) x_1^d$$

If f vanishes on X, then $f(x_0, 0) = 0$ implying that the degree zero homogenous component is also zero leaving us with an extra x_1 term

$$f = x_1 \sum_{d \geq 1} f_d(x_0) x^{d-1} \in \langle x_1 \rangle.$$

Ex: Let $X = \mathcal{V}(x_0 + x_1 - x_2) \subseteq \mathbb{P}^2$ then $\mathcal{I}(X) = \langle x_0 + x_1 - x_2 \rangle$. The ideal into vanishing ideal inclusion is immediate by definition. Now

Definition 1.4. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is a homogenous ideal if for all $f \in I$ every homogenous component of f is also in I.*

In fact we can immediately generate a more useful and straightforward characterization of a homogenous ideal.

Proposition 1.5. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is homogenous if and only if it admits a set of homogenous generators.*

Proof. Suppose that $I = \langle S \rangle$ where S is composed of all homogenous polynomials contained in I. Let $f \in I$, then $f = \sum_{k \geq 0} f_k$. Since $f_k \in \langle S \rangle$ by construction then the sum f must also be contained in $\langle S \rangle$. The other inclusion follows immediately form the fact that S is a subset of the ideal I. Now suppose that I admits a set of homogenous generators, more precisely that $I = \langle S \rangle$ where S is generated by a collection of homogenous polynomials. Let $f \in I$, then by definition

$$f = \sum_{i=1}^{m} g_i h_i$$

where $g_i \in K[x_0, \ldots, x_n]$ and $h_i \in \langle S \rangle$ of homogenous degree d_i for all i. Observe that homogenous components of f of degree k take the form

$$f_k = \sum_{i=1}^{m} (g_i h_i)_k$$

. Note that every monomial in the product $g_i h_i$ is a monomial of g_i multiplied by all of h_i. This means that every monomial of the product is of homogenous degree at least d_i. This ensures that for $k < d_i$, the homogenous component $(g_i h_i)_k = 0$. For $k \geq d_i$, monomials of homogenous degree k must be the product $(g_i)_{k-d_i} h_i$ as every monomial of the product $g_i h_i$ is a product of a monomial of g_i with h_i. This implies that $(g_i h_i)_k \in I$ then f_k is a sum of elements of I implying that it must be contained in I. Therefore I must be a homogenous ideal. ℵ

With this we can give a characterization of not just ideals, but vanishing ideals.

Proposition 1.6. *If $X \subseteq \mathbb{P}^n$ is any subset, then $\mathcal{I}(X)$ is a homogenous radical ideal.*

Proof. We begin by proving that $\mathcal{I}(X)$ is a radical ideal. Suppose there exists an integer m such that $f^m \in \mathcal{I}(X)$, then for a point $a \in X$

$$0 = f^m(a) = (f(a))^m.$$

Since K is an infinite field it has no zero divisors implying that $f(a) = 0$ and that $f \in \mathcal{I}(X)$. Hence $\mathcal{I}(X)$ is a radical ideal. Since f vanishes on a then every homogenous component of f must also vanish on a and is therefore contained in $\mathcal{I}(X)$ implying that it is homogenous. ℵ

With this we go on to establish identical properties to those of affine varieties related to the existence of unique irreducible decompositions of projective varieties. The proofs themselves follow exactly as in the affine case.

1.4 9.4 Affine Restrictions

We saw in the very first section of this chapter that a projective space $\mathbb{P}^n$ could be viewed as $\mathbb{A}^n \sqcup \mathbb{P}^{n-1}$. Quite reasonably we ask whether a projective variety could be stratified in this way. Consider the projective variety $X = \mathcal{V}_{\mathbb{P}}(x_0^2 + x_1^2 - x_2^2) \subseteq \mathbb{P}^2_{\mathbb{C}}$. we saw that the affine component of a projective space is the collection of points where the first coordinate is non-zero or more succinctly when the first coordinate was fixed at 1. Similarly, fix $x_0 = 1$, then we have a subvariety

$$X_0 = \mathcal{V}_{\mathbb{A}}(1 + x_1^2 - x_2^2) \subseteq X$$

whose solutions are entirely contained in $\mathbb{A}^2_{\mathbb{C}}$. We give this phenomenon a more precise definition:

Definition 1.5 (Affine Patch)**.** *For each $i \in \{0, \dots, n\}$ the i-th affine patch of $\mathbb{P}^n$ is the set*

$$\mathbb{A}_i^n = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

and for any set $X \subseteq \mathbb{P}^n$, the intersection $X \cap \mathbb{A}_i^n$ is called the i-th affine restriction of X.

Of course by simply dropping the i-th component, which is fixed at 1, we are in bijection with $\mathbb{A}^n$. Then the i-th restriction of a projective variety $X = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$ is $\mathcal{V}_{\mathbb{A}}(\mathcal{S}_i)$ where

$$\mathcal{S}_i = \{f(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n) \mid f \in \mathcal{S}\}$$

1.5 9.5 Projective Closure

In the previous section we developed a way to take the global projective variety and investigate restriction to local affine patches. Now we aim to develop a method to take some affine patches and join them together into a projective variety. Notice though that in the process of restricting to affine patche lose the points at infinity. So if we were to start with affine patches and attempt to move back into a projective variety, we would need to devise a way to add point at infinity and ensure that the resulting set is actually a projective variety. To do this we define the projective closure.

Definition 1.6 (Projective Closure)**.** *Let $X \subseteq \mathbb{A}^n$ be an affine variety, and let $j_0(X)$ be the image of X in the first affine patch $\mathbb{A}_0^n \subseteq \mathbb{P}^n$. The projective closure of X, denoted $\overline{X}$ is the intersection fo all projective varieties that contain $j_0(X)$.*

Remark 1.3. Observe that, under the zariski topology, this is exactly the definition of a closed set.

- Consider the affine variety $\mathcal{V}_{\mathbb{A}}(1 + x_1 - x_2) \subseteq \mathbb{A}^2$ which has the vanishing set $\{(a, 1+a) : a \in K\}$. Under the josure, j_0 we can embed this variety as a set in $\mathbb{P}^2$ defined as

  $$j_0(X) = \{[1 : a : a + 1] : a \in K\} \subseteq \mathbb{P}^2.$$

  Observe that $[1 : a : a+1] = [\frac{1}{a} : 1 : 1 + \frac{1}{a}]$ means that a must be furthe resitrcted to $a \in K \setminus \{0\}$. Take $b = \frac{1}{a}$ then we can rewrite the elements of $j_0(X)$ to be $[b : 1 : 1 + b]$. Then in terms of our polynomial, for every point in $j_0(X)$, $f(b, 1, 1 + b) = 0$ can be viewed as a univariate polynomial $\tilde{f}(x, 1, 1 + x) = 0$ which means that $\tilde{f}$ is the univariate polynomial. If we were to treate $j_0(X)$ as a projective variety then we have a univariate polynomial vanishing on $\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{P}^1$. In particular this polynomial is vanishing on $\mathbb{P}^1$ which in this case is only the poin $[0 : 1]$ then $f(0, 1, 1)$ must be a solution so we have the point $[0 : 1 : 1]$ as a solution. So we determine that we must augment $j_0(X)$ as $j_0(X) \cup \{[0 : 1 : 1]\}$. Note that we STILL don't yet know whether this is a projective variety because restricting to the first affine path agrees with $j_0(X)$ but we don't know how this variety is changed when restricting to different affine patches. This means that we must determine the polynomial that vanishes and luck for us there is one that I am too lazy to compute so I will take from the book:

  $$\overline{X} = \mathcal{V}_{\mathbb{P}}(x_0 + x_1 - x_2)$$

- Let $\mathcal{V}_{\mathbb{A}}(x_2 + x_1^2) \subseteq \mathbb{A}^2$ has josure

  $$j_0(X) = \{[1 : a : a^2] | a \in K\}$$

  which under $b = \frac{1}{a}$ can be viewed as the set of points $\{[b^2 : b : 1] : b \in K \setminus \{0\}\}$. Similarly we must add a single point and then determine whether this point with the josure is actually the vanishing of the polynomial. It is hence $\overline{X} = \mathcal{V}_{\mathbb{P}}(x_0 x_1 - x_2^2)$.

Observe that the resulting polynomial of the variety is now of some homogenous degree. In fact, as we will state soon, the projective closure of an affine variety is always the "homogenization" of the ideal generating the affine variety. This gives us a much simpler characterization of the projective closure than the rather abstract intersection of all varieties containing the josure.

Definition 1.7. *Let $f \in K[x_1, \dots, x_n]$ be a polynomial of degree d. The homogeniztion of the polynomial is define as*

$$\overline{f} = x_0^d f(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}) \in K[x_0, \dots, x_n].$$

Observe that if we were to look at the josure of a homogenized polynomial, that is we evaluate $\overline{f}(1, x_1, \ldots, x_n)$ we will get back our unhomogenized polynomial f. Reframing our previous examples we would like to determine whether if $X = \mathcal{V}_{\mathbb{A}}(f_1, \ldots, f_k)$ that $\overline{X} = \mathcal{V}_{\mathbb{P}}(\overline{f}_1, \ldots, \overline{f}_k)$ yet the accursed twisted cubic will laugh in our face. I don't feel like walking through this example so I will make some general remarks about it insead. The affine twisted cubic $\mathcal{V}_{\mathbb{A}}(x_2 - x_1^2, x_3 - x_1^3)$ will have a josure $j_0(X) = \{[1 : a : a^2 : a^3] \mid a \in K\} \subseteq \mathbb{P}^3$ and to make it a projective variety we only need to add the point $[0 : 0 : 0 : 1]$. If we were to look at a variety definined by the homogenization of the polynomials then we would arrive at $j_0(X) \sqcup \{0 : 0 : b : c\}$, i.e. we would add too many points. Instead we actually another defining polynomial rather than simply homogenizing the defining polynomials of the affine variety. So instead of homogenizing some polynomials we will actually just homogenize the entire ideal.

Definition 1.8 (Homogenization of a Set)**.** *Let $\mathcal{S} \subset K[x_1, \ldots, x_n]$ be a set of polynomials. The homogenization of S is the set*

$$\overline{\mathcal{S}} = \{\overline{f} : f \in \mathcal{S}\} \subseteq K[x_0, \ldots, x_n]$$

One thing to note is that generally the homogenization of an ideal is not also an ideal. If we were to take a product $g\overline{f}$ where $g \in K[x_0, \ldots, x_n]$ and inhomogenous and $\overline{f} \in \overline{\mathcal{S}}$ then our product is not homogenous and so can not be in $\overline{\mathcal{S}}$. With the homogenization of a set we get the right information to recharacterize the affine closure in a way that plays better with the vanishing and ideal operators.

Proposition 1.7. *If $X \subseteq \mathbb{A}^n$ and affine variety then $\overline{X} = \mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}) \subseteq \mathbb{P}^n$.*

Proof. ($\subseteq$) Let $[1 : a_1 : \cdots : a_n] \in j_0(X)$ and $g \in \overline{\mathcal{I}_{\mathbb{A}}(X)}$ then $g = \overline{f}$ for some $f \in \mathcal{I}_{\mathbb{A}}(X)$ and $g(1, a_1, \ldots, a_n) = f(a_1, \ldots, a_n) = 0$ then $j_0(X)$ is contained in $\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)})$. Since $\overline{X}$ is the smallest projective variety to contain $j_0(X)$ then $\overline{X}$ must be contained in $\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)})$. ($\supseteq$) It suffices to show that

$$\mathcal{I}_{\mathbb{P}}(\overline{X}) \subseteq \mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}))$$

because application of the projective vanishing operator will give us the desired containment. Suppose that $g \in \mathcal{I}_{\mathbb{P}}(\overline{X})$, as g is contained in a homogenous ideal it must also be homogenous. The homogenous polynomial g is not necessarily the homogenization of a polynomial so to force this condition let $k \in \mathbb{N}$ be the maximal choice such that $g = x_0^k h$ for some $h \in K[x_0, \ldots, x_n]$. Observe that forcing k to be maximal implies that from at least one monomial of g we must be able to factor the entirety of an x_0 component out, if this weren't the case then there would be some x_0 terms in every monomial and k would not be maximal. Since g is homogenous of some degree d, then this monomial has degree $d - k$. Note that this monomial must have minimal degree after factoring as any term with some x_0 has less remaining degree distributed over terms x_i where $i \neq 0$. Take $h_0 = h(1, x_1, \ldots, x_n)$ and observe that this removes the x_0 term from every monomial and utilizing the conlusion of the previous two sentences this means that the monomial of degree $d - k$ is now of maximal degree in h_0. The $\overline{h_0} = x_0^{d-k} h_0$ which returns us back to the polynomial h. Since g vanishes on all of $\overline{X}$ then it must vanish on all of $j_0(X)$, in particular this means that g must vanish on all of X. Since

$$h_0(a_1, \ldots, a_n) = h(1, a_1, \ldots, a_n) = g(1, a_1, \ldots, a_n) = 0$$

we see that $h_0 \in \mathcal{I}_{\mathbb{A}}(X)$ and by definition $h \in \overline{\mathcal{I}_{\mathbb{A}}(X)}$ and that $g \in \langle \overline{\mathcal{I}_{\mathbb{A}}(X)} \rangle$. Then

$$g \in \langle \overline{\mathcal{I}_{\mathbb{A}}(X)} \rangle \subseteq \mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}))$$

and we are done. ℵ

While we can't in general conclude that $\overline{\mathcal{V}_{\mathbb{A}}(f_1, \ldots, f_k)} = \mathcal{V}_{\mathbb{P}}(\overline{f_1}, \ldots, \overline{f_k})$ we get a special case when $i = 1$.

Proposition 1.8. *If $f \in K[x_0, \ldots, x_n]$ then $\overline{\mathcal{V}_{\mathbb{A}}(f)} = \mathcal{V}_{\mathbb{P}}(\overline{f})$.*

Proof. Let $X = \mathcal{V}_{\mathbb{A}}(f)$, the Nullstelensatz says that $\mathcal{I}_{\mathbb{A}}(\mathcal{V}_{\mathbb{A}}(X)) = \langle q_1 \cdots q_\ell \rangle$ where $f = q_1^{\alpha_1} \cdots q_\ell^{\alpha_\ell}$ is an irreducible decomposition of f. Using the previous proposition $\overline{\mathcal{V}_{\mathbb{A}}(X)} = \mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}) = \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle})$. It remains to show that this is equivalent to $\mathcal{V}_{\mathbb{P}}(\overline{f})$. Let $f \in \langle q_1 \cdots q_\ell \rangle$ then $\overline{f} \in \overline{\langle q_1 \cdots q_\ell \rangle}$ which implies that

$$\mathcal{V}_{\mathbb{P}}(\overline{f}) \supseteq \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle}).$$

Now suppose that $a \in \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle})$ then for some $\overline{f} \in \overline{\langle q_1 \cdots q_\ell \rangle}$, $\overline{f}(a) = 0$. Note that $\overline{f} = \overline{q_1}^{\alpha_1} \cdots \overline{q_\ell}^{\alpha_\ell}$. The fact that $\overline{f}$ vanishes on a implies that $\overline{q_i}^{\alpha_i}$ vanishes on a for some i. Then for any $g \in \overline{\langle q_1 \cdots q_\ell \rangle}$ it follows that g must also vanish on a because it will have the vanishing homogenized q_i component. ℵ

One thing that we may have noticed in the examples working through projective closures was the fact that we never seemed to be adding affine points, only points at infinity. In fact these are examples of the following proposition.

Proposition 1.9. *The affine restriction of the projective closure of an affine variety X is always X. $(\overline{X})_0 = X$.*

Proof. Let $X \subseteq \mathbb{A}^n$ be an affine variety. Define $\mathcal{S} = \overline{\mathcal{I}_{\mathbb{A}}(X)}$ and note that $\overline{X} = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$. Then the affine restriction $(\overline{X})_0$ vanishes on a set of polynomials

$$\mathcal{S}_0 = \{g(1, a_1, \ldots, a_n) \mid g \in \mathcal{S}\} = \{f(a_1, \ldots, a_n) \mid f \in \mathcal{I}_{\mathbb{A}}(X)\}$$

where the second equality comes from the fact that g must be the homogenization of some polynomial f and evaluating g at $x_0 = 1$ simply returns us to evaluation f at $a_1, \ldots, a_n$. The final set is just $\mathcal{I}_{\mathbb{A}}(X)$ then $\mathcal{V}_{\mathbb{A}}(\mathcal{I}_{\mathbb{A}}(X)) = X$ and we are done. ℵ

1.6 9.6 Project Nullstelensatz

If we are to take inspiration from the affine NS then we aim to prove that

$$\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \sqrt{I}$$

for some ideal $I \in K[x_0, \ldots, x_n]$. But we have a problem here if $I = \langle x_0, x_1, \ldots, x_n \rangle$. The only point that vanishes on a polynomial in here would be $[a_0 : \cdots : a_n]$ where each $a_i = 0$. Since there is no such point in $\mathbb{P}^n$ then $\mathcal{V}_{\mathbb{P}}(I) = \emptyset$. Yet every polynomial vanishes on the emptyset hence $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = K[x_0, \ldots, x_n] \neq I$ where we note that I is already radical. So with this we aim to disregard this annoying ideal.

Definition 1.9. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is irrelivant if $I \supseteq \langle x_0, \ldots, x_n \rangle$. Otherwise I is called relevant.*

With this the projective NS is the affine NS reframed with these classification.

Proposition 1.10. *Let $I \subseteq K[x_0, \ldots, x_n]$ be a homogenous ideal.*

1. *$\mathcal{V}_{\mathbb{P}}(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \langle x_0, \ldots, x_n \rangle$.*
2. *If $\sqrt{I}$ is relevant then $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \sqrt{I}$.*

But before proving this we will utilize a different way to move between affine and projective varieties through cones.

Definition 1.10. *Let $X \subseteq \mathbb{P}^n$ be a subset. The affine cone over X is the set*

$$C(X) = \{(0, \ldots, 0)\} \cup \{(a_0, \ldots, a_n) \mid [a_0 : \cdots : a_n] \in X\} \subseteq \mathbb{A}^{n+1}$$

With the affine cone in mind we'll prove the following proposition that allow interchagnes between the affine and projecive vanishng and ideal operators.

Proposition 1.11. *Let $I \subsetneq K[x_0, \ldots, x_n]$ be a homogenous ideal then*

$$\mathcal{V}_{\mathbb{A}}(I) = C(\mathcal{V}_{\mathbb{P}}(I))$$

Proof. Since we are looking to show that $\mathcal{V}_{\mathbb{A}}(I)$ is a cone then we must first show that for any homogenous ideal that this variety admits $(0, \ldots, 0)$. Suppose that it did not admit the origin, this means that there exsists a homogenous polynomial $f \in I$ such that $f(0, \ldots, 0) \neq 0$. This is only possible if the polynomial is non-zero and constant which forces I to admit such a generator. But if it were to admit such a generator then $I = K[x_0, \ldots, x_n]$ contradicting our choice of I. Since each polynomial in I is homogenous, if it were to vanish on a point $(a_0, \ldots, a_n)$ then it must vanish on any representative $[a_0 : \cdots : a_n]$ and vice versa hence

$$(a_0, \ldots, a_n) \in \mathcal{V}_{\mathbb{A}}(I) \iff [a_0 : \cdots : a_n] \in \mathcal{V}_{\mathbb{P}}(I) \iff (a_0, \ldots, a_n) \in C(\mathcal{V}_{\mathbb{P}}(I))$$

therefore we are done. ℵ

Proposition 1.12. *Let $X \subseteq \mathbb{P}^n$ be a non-empty subset. Then*

$$\mathcal{I}_{\mathbb{P}}(X) = \mathcal{I}_{\mathbb{A}}(C(X)).$$

Proof. Note that $\mathcal{I}_{\mathbb{P}}(X)$ is a non-empty homogenous ideal which means that none of the generating polynomials can be constant. Then on any point $(0,\dots,0) \in \mathbb{A}^{n+1}$ a polynomial in this projective vanishing ideal might vanish. Similarly the fact that a polynomial vanishes on the projective variety X means that it must vanish on it's affine restriction. Then we can redifine the projective vanishing ideal as

$$\mathcal{I}_{\mathbb{P}}(X) = \{f \in K[x_0,\dots,x_n] \mid f(a_0,\dots,a_n) = 0 \text{ if } [a_0 : \cdots : a_n] \in X, f(a_0,\dots,a_n) = 0 \text{ if } (a_0,\dots,a_n) = (0,\dots,0)\} \tag{3}$$

$$= \mathcal{I}_{\mathbb{A}}(\{0,\dots,0\} \cup \{(a_0,\dots,a_n) \mid [a_0 : \cdots : a_n] \in X\}) \tag{4}$$

$$= \mathcal{I}_{\mathbb{A}}(C(X)) \tag{5}$$

ℵ

Now that we have the ability to move between projective and affine varieties and vanishing ideals through cones we can prove the projective NS.

Proof. (a) Suppose that $\sqrt{I} \supseteq \langle x_0,\dots,x_n\rangle$. Note that if $a \in \mathcal{V}_{\mathbb{P}}(I)$ then there exits some polynomial $f \in I$ such that $f(a) = 0$ then $f \in \sqrt{I}$ hence $a \in \mathcal{V}_{\mathbb{P}}(\sqrt{I})$. Suppose that $a \in \mathcal{V}_{\mathbb{P}}(\sqrt{I})$ then for some $m > 0$ there exists an $f^m \in \sqrt{I}$ such that $f^m(a) = 0$. By definition it follows that $f \in I$ and as $f^m(a) = f^{m-1}(a)f(a) = 0$ it follows that $f(a) = 0$. Therefore $\mathcal{V}_{\mathbb{P}}(I) = \mathcal{V}_{\mathbb{P}}(\sqrt{I})$. Then

$$\mathcal{V}_{\mathbb{P}}(I) = \mathcal{V}_{\mathbb{P}}(\sqrt{I}) \subseteq \mathcal{V}_{\mathbb{P}}(\langle x_0,\dots,x_n\rangle) = \emptyset.$$

Now suppose that $\mathcal{V}(I) = \emptyset$, then either $I = K[x_0,\dots,x_n]$ in which case $\sqrt{I} = I \supseteq \langle x_0,\dots,x_n\rangle$. Suppose that $I \subsetneq K[x_0,\dots,x_n]$. Utilizing prop 1.11

$$\mathcal{V}_{\mathbb{A}}(I) = C(\mathcal{V}_{\mathbb{P}}(I)) = C(\emptyset) = \{(0,\dots,0)\}.$$

Under the affine NS

$$\sqrt{I} = \mathcal{I}_{\mathbb{A}}(\mathcal{V}_{\mathbb{A}}(I)) = \mathcal{I}_{\mathbb{A}}(\{0,\dots,0\}) = \langle x_0,\dots,x_n\rangle$$

(b) Now assume that $\sqrt{I}$ is relevant. Suppose that $\mathcal{V}_{\mathbb{P}}(I) = \emptyset$. The first case tells us that $\sqrt{I} \supseteq \langle x_0,\dots,x_n\rangle$ but the only relevant ideal that can contain the maximal irrelevant ideal is the full ring i.e. $\sqrt{I} = K[x_0,\dots,x_n]$. Then $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \mathcal{I}_{\mathbb{P}}(\emptyset) = K[x_0,\dots,x_n]$ and we are done. Now suppose that $\mathcal{V}_{\mathbb{P}}(I) \neq \emptyset$ which means that $I \subsetneq K[x_0,\dots,x_n]$. Then,

$$\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \mathcal{I}_{\mathbb{A}}(C(\mathcal{V}_{\mathbb{P}}(I))) = \mathcal{I}_{\mathbb{A}}(\mathcal{V}_A(I)) = \sqrt{I}$$

hence we are done.

ℵ

Remark 1.4. $\mathcal{V}_{\mathbb{P}}(f_1,\dots,f_k) = \emptyset$ if and only if there exists d such that every monomial of degree d is contained in $\langle f_1,\dots,f_k\rangle$.

Proof. SKETCH : ($\Longleftarrow$) If every monomial of degree d is contained in $\langle f_1,\dots,f_k\rangle$ then in particular we contain $x_0^d,\dots,x_n^d$ on which means that we contain the irrelevant ideal and projective NS tells us this vanishing is empty. ($\Longrightarrow$) If the vanishing is empty then the projective NS tells use that $\sqrt{\langle f_1,\dots,f_k\rangle} \supseteq \langle x_1,\dots,x_n\rangle$. By definition of radical ideals there exist a d_i such that $x_i^{d_i}$ is contained in the radical of the vanishing ideal. Now take $N = \max\{d_1,\dots,d_n\}$ which ensures that any x_i^N is contained in the vanishing ideal. Let $d = (n+1)N$ ℵ

2 CH 10 Maps of Projective Varieties

2.1 10.1 Regular Maps of Projective Varieties

In the affine case we established that polynomials maps between affine varieties maps that are polynomial in each coordinate from one variety into another. In projective varieties this perspective makes less sense because of the scaling factor that acts as an equivalence among projective points. So instead we consider homogenous maps between projective varieties, we have to be careful though because we may accidentally vanish on a point in the variety and end up with a function that isn't well defined. To avoid this we define maps between projective varieties as follows:

Definition 2.1. *Let $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ be projective varieites. A map $F : X \to Y$ is called polynomial if there exists polynomials $f_0, \dots, f_n \in K[x_0, \dots, x_m]$ homogenous of the same degree, such that $X \cap \mathcal{V}(f_0, \dots, f_m) = \emptyset$ and*

$$F(a) = [f_0(a) : \cdots : f_n(a)] \; \forall a \in X$$

If we can take f_i to be linear in all i then we call the map between varieties linear.

Unfortunately the requirement that $X \cap \mathcal{V}(f_0, \dots, f_m)\emptyset$ tends to be far too restrictive. The restrictiveness arises when we try and define polynomial maps over all varieities, this intersection requirement cuts out many obvious maps and as we will see later polynomials maps on projective varieties are always constant. Lets illustrate this with an example, take $X = \mathcal{V}(y^2 - xz) \subseteq \mathbb{P}^2$ and consider the function $x, y \in K[x, y, z]$. We can use these function to define a polynomial map as $X \cap \mathcal{V}(x, y) = \{[0 : 0 : 1]\}$ but we still get a map from a subset of X to $\mathbb{P}^1$ which simply excludes this point:

$$[x : y] : X \setminus \mathcal{V}(x, y) \to \mathbb{P}^1 \tag{6}$$

$$[a : b : c] \mapsto [a : b] \tag{7}$$

Now if we consider polynomials $y, z \in K[x, y, z]$ notice that we again don't get a polynomial map but we get a regular map now that excludes $[1 : 0 : 0]$. Notice that these maps together cover the entirety of X so we can define a map

$$F : X \to \mathbb{P}^1 \tag{8}$$

$$[a : b : c] \mapsto \begin{cases} [a : b], & [a : b : c] \notin \mathcal{V}(x, y) \\ [b : c], & [a : b : c] \notin \mathcal{V}(y, z) \end{cases} \tag{9}$$

In fact for any projective variety X and every point $p \in X$ there exist a pair of possibly different polynomials f, g such that $p \notin \mathcal{V}(f, g)$ meaning we can define a map over the entire projective variety by constructing a piecewise map for every p. This motivates the definition for a regular map:

Definition 2.2. *A map $F : X \to Y$ of projective varieties is called regular if for every $p \in X$, there exists polynomials $f_0, \dots, f_n \in K[x_0, \dots, x_m]$ homogenous of the same degree such that $p \notin \mathcal{V}(f_0, \dots, f_n)$ and*

$$F(a) = [f_0(a) : \cdots : f_n(a)] \; \forall a \in X \setminus \mathcal{V}(f_0, \dots, f_n)$$

It's important to note that every polynomial map is a regular map, in the sense that we were able to find a single polynomial for all points in the projective variety, but not every regular map is polynomial.

Remark 2.1. Any regular map $F : \mathbb{P}^1 \to \mathbb{P}^n$ is a polynomial map.

Proof. Let F be a regular map from $\mathbb{P}^1$ to $\mathbb{P}^n$. This implies that for any point $p \in \mathbb{P}^1$ there exists polynomial $f_0, \dots, f_n \in K[x, y]$ homogenous of the same degree such that $p \notin \mathcal{V}(f_0, \dots, f_n)$ and that

$$F(a) = [f_0(a) : \cdots : f_n(a)]$$

for all $a \in \mathbb{P}^1 \setminus \mathcal{V}(f_0, \dots, f_n)$. If $\mathbb{P}^1 \cap \mathcal{V}(f_0, \dots, f_n) = \emptyset$ then we are done. Assume that the intersection is non-empty then there exists a point $p \in \mathbb{P}^1$ such that $f_i(p) = 0$ for all i. If $p = [0 : a]$ for some $a \in K \setminus \{0\}$ then we may view each f_i as a univariate polynomial vanishing at a. Since our field is

algebraically closed then $f_i(x,y) = (y-a)g_i(x,y)$ allowing us to choose a new collection $g_0, \ldots, g_n$ which doesn't vanish at p, is equivalent upon evaluation as point in $\mathbb{P}^n$, and that only add p to the domain of F. If $p = [1 : a]$ for some $a \in K$, then we can do the same process to product a collection on which this point doesn't vanish. Once we have a collection that no longer vanishes at p, we repeat the process by checking for projective points on which this new collection doesn't vanish. Note that any polynomial only has a finite number of solutions ensureing that this process terminates at some collection $h_0, \ldots, h_n$. In particular this process must terminate a point where no polynomial has a common root or we have forced a 1 in some component in both cases implying the vanishing set is empty. ℵ

2.2 10.2 Projective Equivalences

Of course now that we have developed the notion of a map between projective varieties. This naturally asks us to consider when two projective varieties are isomorphic to one another. In particular, if for projective varieties $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ there exist a pair of regular maps $F : X \rightleftarrows Y : G$ where $F \circ G = \mathrm{id}_X$ then we say these varieites are isomorphic. Verifying an isomorphism has no special conditions other than ensuring that given a map F, that the inverse that you define is a well defined regular map on Y. We are particularly interested in a very special isomorphism of $\mathbb{P}^n$ to $\mathbb{P}^n$. Similar to affine space we can identify $\mathbb{P}^n$ with a vector space, note that we could never really be a vector space because we never have the origin, and consider for a point $[a_0 : a_1 : a_2] \in \mathbb{P}^2$

$$\begin{bmatrix} a_0 & a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

giving us a linear map

$$[a_0 : a_1 : a_2] \mapsto [a_0 + a_1 : a_0 + 2a_2 : a_1 - a_2].$$

More generally for a point $[a_0 : \cdots : a_n]$ and a $(n+1) \times (n+1)$ invertible matrix we get a linear map

$$F_A = \left[\sum_{j=0}^{n} A_{0,j} x_j : \cdots : \sum_{j=0}^{n} A_{n,j} x_j \right]$$

that, given the invertibility of A, has an inverse $F_{A^{-1}}$. This then implies that F_A and $F_{A^{-1}}$ are enough information to provide an isomorphism of projective n space to itself.

Definition 2.3 (Projective Equivalence)**.** *A projective equivalence of $\mathbb{P}^n$ is an isomorphism of the form*

$$F_A : \mathbb{P}^n \to \mathbb{P}^n \tag{10}$$

$$p \mapsto Ap \tag{11}$$

where A is an $(n+1) \times (n+1)$ matrix with entries in K. We say that projective varieties $X, Y \subseteq \mathbb{P}^n$ are projectively equivalent if there exists a projectively equivalent if there exists a projective equivalence such that $F_A(X) = Y$.

Remark 2.2. If X is a projective variety then a projective equivalence always maps $X = \mathcal{V}(f_1, \ldots, f_k)$ to a projective variety $F_A(X)$. To see this consider that

$$F_A(X) = \{F(a) \in \mathbb{P}^n : f_i(a) = 0 \ \forall i\} = \{a \in \mathbb{P}^n : (f_i \circ F_A^{-1})(a) = 0 \ \forall i\}$$

where $f_i \circ F_A^{-1}$ is polynomial because it is just a composition of two polynomial. Then it's immediate to verify that $(f_i \circ F_A^{-1})(F_A(a)) = f_i(a) = 0$. This assertion is generally true but requires us to develop some more machinery and will be tackled in Theorem 11.65.

The language of projective equivalences, on a degree two polynomial, exhibits a remarkable level of regularity. That is

Theorem 2.3. *Let $f, g \in K[x, y, z]$ be irreducible and homogenous of degree 2. Then $\mathcal{V}(f)$ and $\mathcal{V}(g)$ are projectively equivalent in $\mathbb{P}^2$.*

Proof. Choose two distinct points $[a_1 : b_1 : c_1], [a_2 : b_2 : c_2] \in \mathcal{V}(f)$, and the matrix

$$\begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \\ c_1 & c_2 \end{bmatrix}$$

must be of rank 2. If every point of $\mathcal{V}(f)$ satisfies the linear equations given by this matrix then we have $\mathcal{V}(f)$ to be contained in the vanishing set of a linear function. But the radical of the ideal generated by a linear polynomial is just the same ideal which implies that we were able to express a quadratic of homogenous degree 2 as the product of a linear polynomial. This contradiction implies that there must exist a third point $[a_3 : b_3 : c_3]$ such that

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

is off full rank and therefore invertible. Therefore A induces a projective equivalence F_A defined by A maps $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]$ to our three points chosen from $\mathcal{V}(f)$. The inverse identifies $\mathcal{V}(f)$ with some irreducible conic $\mathcal{V}(g)$ such that those three points are contained in said irreducible conic. But a homogenous polynomial of degree 2 that vanishes on those three points has the form

$$g = \alpha xy + \beta yz + \gamma xz.$$

All α, β, and γ must be non-zero or g would be reducible contradiction our assertion that $\mathcal{V}(g)$ is irreducible. Then the projective equivalence must be a map

$$[a : b : c] \mapsto [\alpha^{-1}a : \beta^{-1}b : \gamma^{-1}c]$$

which identifies $\mathcal{V}(g)$ with projective variety $\mathcal{V}(xy + xz + yz)$. Then composing these we see that we have a projective equivalence. ℵ

From this we can actually make a rather powerful isomorphism statememt:

Theorem 2.4. *Let $f \in K[x, y, z]$ be an irreducible, homogenous polynomial of degree 2. Then $\mathcal{V}(f) \cong \mathbb{P}^1$.*

Becuase we know that all varieties of this form are isomorphic to one another it just suffices to show that one of these is isomorphic to $\mathbb{P}^1$. Refering to $\mathcal{V}(y^2 - xz)$ we can show this and the result of the proof follows.

2.3 10.3 Veronese Maps

Veronese maps are special because they give us a mechanism to identify, under projective equivalence, any variety defined by complex polynomials of high homogenous degree with varieties defined by a collection of polynomials of degree at most two. Using this result we can then establish that any polynomials map of projective varieties is linear(we saw something similar to this argument in 10.1.6).

Definition 2.4 (Veronese Maps)**.** *For any $d, n \geq 1$, the Veronese map $F_{d,n}\mathbb{P}^n \to \mathbb{P}^{\binom{d+n}{k}-1}$ is the polynomial map*

$$F_{d,n} = [x_0^d : x_0^{d-1}x_1 : x_0^{d-1}x_2 : \cdots : x_n^d]$$

where the monomials appearing in the definition are all of the possible monomials of degree d in the variables $x_0, \ldots, x_n$.

Note that we can view $F_{d,n}$ as a regular map because of the manner in which we defined it. That is let $\Delta = \mathcal{V}(x_0^d, x_0^{d-1}x_1, x_0^{d-1}x_2, \ldots, x_n^d)$ and observe that if a point $p \in \mathbb{P}^n \cap \Delta$ then the componentes $p_i^d = 0$ for all i which means that all $p_i = 0$ but a point with zeroes in each component can not be contained in $\mathbb{P}^n$. Since $F_{d,n}$ is polynomial it is by definition regular. Let $D = (d_0, \ldots, d_n)$ be the tuples which index the degree $d_0 + \cdots + d_n = d$ monomials. For example

$$X^D = x_0^{d_0} \cdots x_n^{d_n}$$

and we will call the cordinates of $\mathbb{P}^{\binom{d+n}{k}-1}$ by y_D so that $F_{d,n}$ is the polynomial map whose y_D-coordinate is x^D.

Proposition 2.1. *For any $d, n \geq 1$, the image of the Veronese map $F_{d,n} : \mathbb{P}^n \to \mathbb{P}^{\binom{d+n}{k}-1}$ is*

$$X_{d,n} = \mathcal{V}(\{y_D y_{D'} - y_E y_{E'} \mid D + D' = E + E'\}) \subseteq \mathbb{P}^{\binom{d+n}{k}-1}$$

Proof. Let $a \in F_{n,d}(\mathbb{P}^n)$ then for any D, D', E, E' such that $D + D' = E + E'$ given by the vanishing

$$a^D a^{D'} - a^E a^{E'} = (a_0^{d_0} \cdots a_n^{d_n})(a_0^{d'_0} \cdots a_n^{d'_n}) - (a_0^{e_0} \cdots a_n^{e_n})(a_0^{e'_0} \cdots a_n^{e'_n}) = 0$$

then we see immediately that $F_{d,n}(\mathbb{P}^n) \subseteq X_{d,n}$. Now take $a \in X_{d,n}$ then it follows by the conditions on the vanishing that for tuples D, D', E, E' that $a_D a_{D'} = a_E a_{E'}$. This can be immediately extended to show that for any D_k, E_k where $D_1 + \cdots + D_k = E_1 + \cdots + E_k$ that

$$a_{D_1} \cdots a_{D_k} = a_{E_1} \cdots a_{E_k}.$$

In particular for $D = (d_0, \ldots, d_n)$ we have that

$$a_D^d = (a_0^{d_0} \cdots a_n^{d_n})^d = a_{(d,0,\ldots,0)}^{d_0} \cdots a_{(0,\ldots,d)}^{d_n}.$$

Note that at least one coordinate must be non-zero because each index a_i comes from a component of a projective class in $\mathbb{P}^n$. WLOG suppose that $a_{(d,\ldots,0)} \neq 0$ then we can rescale this projective component by this component so that $a_{(d,\ldots,0)} = 1$. For $1 \leq i \leq n$ define

$$b_i = b_{((d-1),\ldots,1,\ldots,0)}$$

where the 1 lives in the i-th index of the tuple. Then

$$\begin{aligned}
F_{d,n}(b) &= b_0^{d_1} \cdots b_n^{d_n} && (12)\\
&= a_{(d,0,\ldots,0)}^{d_0} a_{(d-1,1,\ldots,0)}^{d_1} \cdots a_{(d-1,0,\ldots,1)}^{d_n} && (13)\\
&= (a_0^{d-1} a_1)^{d_1} \cdots (a_0^{d-1} a_n)^{d_n} && (14)\\
&= (a_0)^{(d-1)d_1} \cdots a_0^{(d-1)d_n} a_D && (15)\\
&= a_0^{(d-1)d} a_D && (16)\\
&= a_{(d,\ldots,0)}^{d-1} a_D && (17)\\
&= a_D && (18)
\end{aligned}$$

where (14) and (18) are a result of the fact that we set $a_0 = 1$. Therefore $b \in \mathit{image}(F_{d,n}(\mathbb{P}^n))$ and $X_{d,n} \subseteq F_{d,n}(\mathbb{P}^n)$. ℵ

Whats good is that we get explicit defining polynomials but we can even show that Veronese maps are in fact isomorphism of $X_{d,n}$ and $\mathbb{P}^n$.

Proposition 2.2. *For any $d, n \geq 1$, the Veronese map $F_{d,n} : \mathbb{P}^n \to X_{d,n}$ is an isomorphism.*

Proof. To prove that this is an isomorphism we construct the following map for any $b \in X_{d,n}$ define

$$G_{d,n}(b) = \begin{cases} [b_{(d,0,\ldots,0)} : b_{(d-1,1\ldots,0)} : \cdots : b_{(d,0,\ldots,1)}] \\ [b_{(1,d-1,\ldots,0)} : b_{(0,d,\ldots,0)} : \cdots : b_{(0,d-1,\ldots,1)}] \\ \qquad \vdots \\ [b_{(1,0,\ldots,d-1)} : b_{(0,1,\ldots,d-1)} : \cdots : b_{(d,0,\ldots,d)}] \end{cases}$$

where we choose the correct presentation depending on the point b. In the previous proof we saw that not every component could be 0 so we know that the point given by G is well defined. It remains to check the composition but that seems tedious and I am happy to accept that this is an inverse. ℵ

Ex: Let $X = \mathcal{V}(x_0^3 + x_1^2 + x_2^3) \subseteq \mathbb{P}^2$ and we consider the Veronese map

$$F_{3,2} : X \to \mathbb{P}^{\binom{5}{3}-1} = \mathbb{P}^9$$

Observe that $D = (d_0, d_1, d_2)$ is the tuple indexing outputs of the Veronese map and $d_1+d_2+d_3 = 3$. Then we can verify that

$$a \in X \iff a_0^3+a_1^3+a_2^3 = 0 \iff a_{(3,0,0)}+a_{(0,3,0)}+a_{(0,0,3)} = 0 \iff F_{3,2}(a) \in \mathcal{V}(y_{(3,0,0)}, y_{(0,3,0)}, y_{(0,0,3)})$$

then

$$F_{3,2}(X) \cong X_{3,2} \cap \mathcal{V}(y_{(3,0,0)}, y_{(0,3,0)}, y_{(0,0,3)})$$

following from the previous proposition. More generally we can present the first most important consequences of the relationship given by Veronese maps.

Proposition 2.3. *Up to isomorphism, every projective variety can be written as the vanishing set of a finite collection of homogenous polynomials of degree at most two.*

Proof. Let $X \subseteq \mathbb{P}^n$ be a projective variety. The primary observation is that X can be expressed as the vanishing set of polynomials $f_1, \dots, f_k$ of the same homogenous degree. Then each f_i has the form

$$f_i = \sum_D a_{i,D} x_0^{d_0} \cdots x_n^{d_n}$$

over all tuples $D = (d_0, \dots, d_n)$ *verify this claim*. Now consider the Veronese map

$$F_{d,n} : \mathbb{P}^n \to \mathbb{P}^{\binom{n+d}{d}-1}$$

and define linear polynomials $\ell_1, \dots, \ell_k$ such that

$$\ell_i = \sum_D a_{i,D} y_D$$

We are trying to get a Veronese map whose image is a variety which lets us treat the map as an ismorphism Then we observe that

$$a \in X \iff F_{d,n}(a) \in X_{d,n} \cap \mathcal{V}(\ell_1, \dots, \ell_k)$$

which implies that the image of the regular map on the variety X is exactly this intersection. Since the map is regular it can be restricted to a regular map on X. But because we know that the Veronese map is an isomorphism onto the variety $X_{d,n}$ then we have a regular invese to the map giving us an isomorphism $X \cong X_{d,n} \cap \mathcal{V}(\ell_1, \dots, \ell_k)$ where everthing defining this isomorphic variety is given by quadratics and linear polynomials. ℵ

This result is particularly interesting because we can say that up to isomorphism, polynomial maps between variety are linear.

Proposition 2.4. *If $F : X \to Y$ a polynomial map of projective varieties, then there exists an isomorphism $G : X \to Z$ and a linear map $L : Z \to Y$ such that $F = L \circ G$.*

Proof. *Exercises - in HW5* ℵ

With this and exercise 10.2.6 we can impose a rather strong condition on polynomial maps.

Proposition 2.5. *A $F : X \to Y$ is a polynomial map of projective varieties, then for any $b \in Y$, there are finitely many $a \in X$ such that $F(a) = b$.*

Proof. Using the previous proposition, $F = L \circ G$ for some linear map L and isomorphism G. Because isomorphisms are one-to-one and linear maps are finite-to-one then F must be finite-to-one. ℵ

2.4 10.4 Segre Maps

We have a solid understanding of projective varieties and ways in which we can embed them into higher dimensional spaces. Now what if have $\mathbb{P}^m \times \mathbb{P}^n$, with our affine intuition we wonder: Does this naturally live in $\mathbb{P}^{m+n}$? The way in which I interpret this is to think about a term $a \in \mathbb{P}^{m+n}$ where the first m terms are zero. If we were to break this back down into the product structure this would leave us with an all zero point in $\mathbb{P}^m$ which doesn't make sense. Fortunately for us, Segre has devised a way to embed this product into a higher projective space via a Segre map.

Definition 2.5 (Segre Map)**.** *For $m, n \geq 0$ the Segre map $S_{m,n} : \mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^{(m+1)(n+1)-1}$ is a map*

$$S_{m,n} = [x_0y_0 : x_0y_1 : \cdots : x_my_{n-1} : x_my_n]$$

where the monomials are all possible products x_iy_j.

Observe that if all products $a_ib_j = 0$ take $b_j = 0$ then for all i, $a_ib_j = 0$ which means that a_i is not necessarily zero for all i. Because we have any other term a_ib_k and we take $a_i \neq 0$ then we force all b_j to be zero. So we only get zero if $a_i = 0$ for all i or $b_i = 0$ for all i. Because this can't happen in $\mathbb{P}^n$ or $\mathbb{P}^m$ then we are good and the Segre map is well defined. Lets look at a simple example

$$S_{1,1} : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3 \tag{19}$$

$$([a_0 : a_1], [b_0 : b_1]) \mapsto [a_0b_0 : a_0b_1 : a_1b_0 : a_1b_1] \tag{20}$$

Observe that in general we can organize entries in $\mathbb{P}^{(m+1)(n+1)-1}$ into components z_{ij} where $0 \leq i \leq m$ and $0 \leq j \leq n$. These entries can be used to populate a matrix $A(z)$. Using our previous example we take $z = [a_0b_0 : a_0b_1 : a_1b_0 : a_1b_1]$

$$A(z) = \begin{bmatrix} a_0b_0 & a_0b_1 \\ a_1b_0 & a_1b_1 \end{bmatrix}$$

Using this presentation we can give the cartesian product of projective spaces by embedding them as the vanishing set in a higher dimensional projective space using a Segre map.

Proposition 2.6. *$S_{m,n}$ is an injection of $\mathbb{P}^m \times \mathbb{P}^n$ as a projective variety $Z_{m,n} \subseteq \mathbb{P}^{(m+1)(n+1)-1}$ where*

$$Z_{m,n} = \mathcal{V}(\{z_{ij}z_{k\ell} - z_{i\ell}z_{kj} : 0 \leq i, k \leq m, 0 \leq j, \ell \leq n\}).$$

Proof. We will begin by showing that $\operatorname{im}(S_{m,n}) = Z_{m,n}$. Take $(a, b) \in \mathbb{P}^m \times \mathbb{P}^n$ and take $z = S_{m,n}(a, b)$ then for $0 \leq i, k \leq m$ and $0 \leq j, \ell \leq n$ we see that

$$z_{ij}z_{k\ell} - z_{i\ell}z_{kj} = a_ib_ja_kb_\ell - a_ib_\ell a_kb_j = 0$$

hence $\operatorname{im} S_{m,n} \subseteq Z_{m,n}$. For $c \in Z_{m,n}$ implying that all 2×2 minors of $A(c)$ vanish. Then $\operatorname{rk} A(c) = 1$ because if the rank was zero this would imply that c was zero in every component. Since $\operatorname{rk} A(c) = 1$ then every row is a multiple of a single row. Let this single row be $b = (b_0, \ldots, b_m)$ and take every other row to be a_ib for some $a_i \in K$. Then we see that

$$c = S_{m,n}([1 : a_1 : \cdots : a_m], [b_0 : \cdots : b_n])$$

hence $\operatorname{im}(S_{m,n}) = Z_{m,n}$. Now we show that $S_{m,n}$ is injective. Consider $z, z' \in Z_{m,n}$ such that $z = S_{m,n}(a, b)$ and $z' = S_{m.n}(x, y)$ for some some $a, x \in \mathbb{P}^m$ and $b, y \in \mathbb{P}^n$ where $z = z'$. This implies that $A(z) = A(z')$, and as we saw in the previous part of this proof, the fact that each 2×2 minor vanishes in both matrices means that they must be rank 1. In particular this means that the first row of $A(z)$ can be set to b and each successive row set to a_ib. Similarly the first row of $A(z')$ can be set to y and the successive rows are multiples x_iy. Since these matrices are equivalent this means that $b = y$. Then $a_ib = x_iy = x_ib$ which implies that $x_i = a_i$ for all i. Therefore $(a, b) = (x, y)$ and we can conlcude that the Segre map is injective. ℵ

Now that we can view a product of the projective spaces as varieties, does this mean that taking products of the varieties of these spaces can be injected to a variety of a higher projective dimension.

Proposition 2.7. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective vareities, then the Segre map $S_{m,n}$ resitrcts an injective variety $X \times Y$ onto a projective variety in $\mathbb{P}^{\binom{n+d}{d}-1}$.*

Proof. Let $X = \mathcal{V}(\mathcal{S})$ and $Y = \mathcal{V}(\mathcal{T})$ be projetive varieties for polynomials $\mathcal{S} \subseteq K[x_0, \ldots, x_m]$ and $\mathcal{T} \subseteq K[x_0, \ldots, x_n]$. For all $f \in \mathcal{S}$ and $g \in \mathcal{T}$ define

$$f_i(z) = f(z_{0i}, \ldots, z_{mi}) \quad \text{and} \quad g_j(z_{j0}, \ldots, z_{jn})$$

collecting all $f_i \in \mathcal{S}'$ and $g_j \in \mathcal{T}'$. Then for $(x, y) \in X \times Y$ we know that $S_{m,n}(x, y) \in Z_{m,n}$ and observe that for all i

$$f(x_0 y_i, \ldots, x_m y_i) = y_i^d f(x_0, \ldots, x_m) = 0$$

and

$$g(x_j y_0, \ldots, x_j y_n) = x_j^d g(y_0, \ldots, y_m)$$

then $S_{m,n}(X \times Y) \subseteq Z_{m,n} \cap \mathcal{V}(\mathcal{S}' \cup \mathcal{T}')$. Reversing this process we can check again that vanishing on $\mathcal{S}', \mathcal{T}'$ will imply that the original points vanish on $\mathcal{S}$ and $\mathcal{T}$. ℵ

With this we can define the product of projective varieties.

Definition 2.6. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective varieties then their product $X \times Y$ is the projective variety*

$$S_{m,n}(X \times Y) \subseteq \mathbb{P}^{(m+1)(n+1)-1}$$

Now we ask whether maps into or out of products of varieities are regular. In particular, our soon to be topological perspecitve, wants us to determine whether the projection maps of $X \times Y$ are regular.

Proposition 2.8. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective varieites. Then the projection maps $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are regular.*

Under the Segre map we may note that for z we can always find m terms $z_{i,0}, \ldots, z_{i,m}$ which when mapped into Y gives us a point scaled by x_j which is equivalent to a point in X.

Proof. We construct π_1 and π_2 follows immdietly from symmetry. For $c = (a, b) \in X \times Y \subseteq \mathbb{P}^{(m+1)(n+1)-1}$ define

$$\pi_1(c) = \begin{cases} [c_{00} : c_{10} : \cdots : c_{m0}] & c \notin \mathcal{V}(z_{00}, \ldots, z_{m0}) \\ \vdots & \\ [c_{0n} : c_{1n} : \cdots : c_{mn}] & c \notin \mathcal{V}(z_{0n}, \ldots, z_{mn}) \end{cases}$$

For $c = (a, b) \in X \times Y$, at least one a_i or b_j is non-zero. Without loss of generality take $b_j \neq 0$ then $c \notin \mathcal{V}(z_{0j}, \ldots, z_{nj})$ and

$$\pi_1(c) = [a_0 b_j : \cdots : a_n b_j] = [a_0 : \cdots : a_n].$$

ℵ

We've established that we can essentially map out of a product of projective varieties. Now we consider maps into a product of varieties.

Proposition 2.9. *Let $W \subseteq \mathbb{P}^\ell$, $X \subseteq \mathbb{P}^m$, and $Y \subseteq \mathbb{P}^n$ be projective varieties. If $F : W \to X$ and $G : W \to Y$ are regular maps, then the function*

$$F \times G : W \to X \times Y \tag{21}$$

$$a \mapsto (F(a), G(a)) \tag{22}$$

Proof. Since F and G are regular for a point $p \in W$ there exist polynomials

$$f_0, \ldots, f_m, g_0, \ldots, g_n \in K[w_0, \ldots, w_\ell]$$

such that

$$F(a) = [f_0(a) : \cdots : f_m(a)] \; \forall a \in W \setminus \mathcal{V}(f_0, \ldots, f_m)$$

and

$$G(a) = [g_0(a) : \cdots : g_n(a)] \; \forall a \in W \setminus \mathcal{V}(g_0, \ldots, g_n).$$

Then by definition

$$p \notin \Delta = \mathcal{V}(f_i g_j : 0 \leq i \leq m, 0 \leq j \leq n)$$

then

$$(F \times G)(a) = [f_0(a)g_1(a) : f_0 g_2(a) : \cdots : f_m(a)g_n(a)]$$

for all $a \in W \setminus \Delta$. This means that by definition the product of regular maps is regular. ℵ

This result is fairly interesting because it says that varieties that arise as products have intrinsic nature dependent on each of their components.

Proposition 2.10. *Let X, X', Y, Y' be projective varieties. If $X \cong X'$ and $Y \cong Y'$, then*

$$X \times Y \cong X' \times Y'$$

Proof. We've established that the projection maps are regular so we consider their precomposition

$$F \circ \pi_1 : X \times Y \to X' \qquad G \circ \pi_2 : X \times Y \to Y'$$

are regular maps and their product is a regular map

$$X \times Y \to X' \times Y'$$

Because F and G are isomorphisms then they have regular inverses and we can repeat this process for F^{-1} and G^{-1} to product an inverse map. Since we have regular maps with regular inverses then we can conclude

$$X \times Y \cong X' \times Y'$$

ℵ

Proposition 2.11. *Let Z be a subset of $\mathbb{P}^m \times \mathbb{P}^n$. Then viewing the product as a projective variety via the Segre map, Z is itself a projective variety if and only if Z is a vanishing set $\mathcal{V}(\mathcal{S}) \subseteq \mathbb{P}^m \times \mathbb{P}^n$ for some $S \subseteq K[x_0, \ldots, x_m, y_0, \ldots, y_n]$.*

Before proving this proposition we will prove a collection of small lemmas beforehand then utilize those lemmas.

Lemma 2.1. *Every polynomial $f \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ decomposes uniquely into a sum of bihomogenous polynomials.*

Proof. *Worth looking at this again just to nail the communication - there are some kind of annoying wording bits that makes it difficult to tell if this is really clear.* Let $f \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ be a homogeneous polynomial of degree d. Observe that monomials of f take the form

$$x_0^{\alpha_0} \cdots x_m^{\alpha_m} y_0^{\beta_0} \cdots y_n^{\beta_n}$$

Suppose that $m \leq n$ let $D_\alpha = (\alpha_0, \ldots, \alpha_m, 0, \ldots, 0)$ and $D_\beta = (\beta_0, \ldots, \beta_n)$ with enough zeroes in D_α such that $D_\alpha, D_\beta \in \mathbb{N}^{n+1}$. Then the above monomial may be viewed as $x_{D_\alpha} y_{D_\beta}$ which we establish in exercises 10.3.3 to be uniquely determines by $D_\alpha + D_\beta$. Let $f_{i,d-i}$ be the sum of monomials in f such that $\sum_{k=0}^m \alpha_k = i$ and $\sum_{k=0} \beta_k = d - i$ and observe that

$$f = \sum_{i=0}^{d} f_{i,d-i}$$

is the unique decomposition of f as each monomial is determined by the D_α and D_β associated to them. Now take a polynomial $g \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ not necessarily homogeneous. The polynomial g can be further decomposed as a sum $\sum_{j=0}^n g_j$ where g_j is the collection of monomials of g of homogeneou degree j. Each homogeneous component g_j can be further decomposed as a sum giving us a decomposition

$$g = \sum_{j=0}^{n} \sum_{i=0}^{d_j} g_{i,d_j-i}$$

Since each homogeneous component decomposes uniquely, then we guarantee that g can be decomposed into a unique sum of bihomogeneous polynomials. ℵ

Lemma 2.2. *Let $(a, b) \in \mathbb{P}^m \times \mathbb{P}^n$ and $f \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ then $f(a, b) = 0$ if and only if every bihomogeneous component of f vanishes at (a, b).*

Proof. Suppose that $f \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ vanishes at a point $(a, b) \in \mathbb{P}^m \times \mathbb{P}^n$. The polynomial f can be decomposed into homogenous components f_k which must vanish on (a, b) for all k. Each f_k can be further decomposed into a sum

$$f_k(a, b) = \sum_{i=0}^{d_k} f_{i,d_k-i}(a, b) = 0.$$

As each f_k vanishes on all representatives of a and b, take $\lambda \in K \setminus \{0\}$ then

$$f_k(\lambda a, \lambda b) = \sum_{i=0}^{d_k} \lambda^{d_k} f_{i,d_k-i}(a, b) = 0$$

We may view this polynomial as a univariate polynomial

$$\sum_{i=0}^{d_k} x^{d_k} f_{i,d_k-i}(a, b) \in K[x]$$

vanishing on infinitely many x. Since we are over an infinite field this implies that this univariate polynomial is a zero polynomial and as a result $f_{i,d_k-i}(a, b) = 0$ for all i. Therefore we can conclude that if $f(a, b) = 0$ then every bihomogenous component of f must vanish on (a, b). Now suppose that every bihomogenous component of f vanish on (a, b). Because each homogenous component of f can be decomposed into a sum of bihomogeneous components, all of which vanish on (a, b) then they must all vanish on (a, b). This means exactly that f vanishes on (a, b). *Do I need to check that bihomogeneous components vanish on all representatives. Probably a good idea just to be careful* ℵ

Finally we get the last and most important result of this seciton that will let us truly think about varieties in products of projective spaces as true varieties.

Proposition 2.12. *Let $Z \subseteq \mathbb{P}^m \times \mathbb{P}^n$ be a subset. Then, viewing $\mathbb{P}^m \times \mathbb{P}^n$ as a projective variety via the Segre map, Z is itself a projective variety if and only if Z is a vanising set $\mathcal{V}(\mathcal{S}) \subseteq \mathbb{P}^m \times \mathbb{P}^n$ for some $\mathcal{S} \subseteq K[x_0, \ldots, x_m, y_0, \ldots, y_n]$.*

2.5 10.5 Grassmannians and Plucker Maps

Grassmanians are special sets $G(n, k)$ that collect all n-dimensional subspaces of k-dimensional vector spaces. Remember back to the first section where we identify projective spaces as line through the origin. Precisely the one dimensional subspaces of K.

Definition 2.7. *A subset $L \subseteq \mathbb{P}^n$ called a k-dimensional linear subvariety of $\mathbb{P}^n$ if there exists a $(k+1)$-dimensional linear subspace $V \subseteq K^{n+1}$ such that*

$$L = \mathbb{P}(V) = \{[a_0 : \cdots : a_n] : (a_0, \ldots, a_n) \in V \setminus \{0\}\}$$

Looking at this definition, observe that the affine cone $C(V)$ is exactly the linear subvariety V. In fact it follows that k-dimensional subvarieties of $\mathbb{P}^n$ are in bijection with $(k+1)$-dimensional linear subspaces of K^{n+1}. This allows us to give a characterization of the grassmanian in terms of projective space.

Definition 2.8. *Let k and n be positive integers with $k \leq n$. The Grassmanian $\mathbb{G}(k, n)$ is the set of all $(k-1)$-dimensional linear subvarieties of $\mathbb{P}^{n-1}$.*

Our goal is to give the Grassmanian structures as a subvariety of projective space. To do this we will examine, for each $V \in \mathbb{G}(n, k)$, the matrix reprsentation of V w.r.p to the basis vectors. Suppose that $V \subseteq K^4$ be the span of $v_1 = (2, 1, -3, 0)$ and $v_2 = (1, 0, 1, 1)$. Since they are L.I. they form a basis of V with a matrix representation

$$\begin{bmatrix} 2 & 1 & -3 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

and suppose that we have an alternate representative given by a different span of linearly independent vectors:

$$\begin{bmatrix} 3 & 2 & -7 & -1 \\ 3 & 1 & -2 & 1 \end{bmatrix}$$

Observe that these representations are different but one can be recovered through the other via the change of basis matrix

$$\begin{bmatrix} 2 & 1 & -3 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 & -7 & -1 \\ 3 & 1 & -2 & 1 \end{bmatrix}$$

Observe that any determinant of 2×2 minors of the left matrix is a product of the determinant of a 2×2 minor in the corresponding columns of the other representation scaled by the determinant of the change of basis matrix. In naive language we are almost saying that one representation is the same as another up to scaling, hinting to us that there might be some projective shenanigans gonig on. To make this more precisefor a matrix M let $I \subseteq \{1, \ldots, n\}$ such M_I is the $k \times |I|$ submatrix of M consisting of columns indexed by I. With this we can define the Plucker map

Definition 2.9. *The Plucker map is a function*

$$P_{k,n} : \mathbb{G}(k, n) \to \mathbb{P}^{\binom{n}{k}+1}$$

where a subspace $V \in \mathbb{G}(k, n)$, with $M(V)$ being the $k \times n$ matrix representatino of V with basis elements as rows, is mapped to the value $P_{k,n}(V) \subseteq \mathbb{P}^{\binom{n}{k}-1}$ in the homogeneous coordinates index by I to $\det(M(V)_I)$.

Going back to our previous example we have the following indices ordered lexicographically

$$x_{12} \quad x_{13} \quad x_{14} \quad x_{23} \quad x_{24} \quad x_{34}$$

where x_{12} corresponds to columns 1 and 2 of the give matirix representation, then computing determinants

$$x_{12} = \det \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = -1 \qquad x_{13} = \det \begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} = 5$$

$$x_{14} = \det \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = 2 \qquad x_{23} = \det \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} = 1$$

$$x_{24} = \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \qquad x_{34} = \det \begin{bmatrix} -3 & 1 \\ 0 & 1 \end{bmatrix} = -3$$

giving us projective point $[-1 : 5 : 2 : 1 : 1 : -3]$. Similarly we can compute for the second representation giving us the same point $[-3 : 15 : 6 : 3 : 3 : -9]$ as its a scalling of the other point.

Proposition 2.13. *For any integer k and n with $0 \leq k \leq n$, the Plucker map $P_{k,n}$ is injective.*

Proof. Let $a \in \operatorname{im}(P_{k,n}) \subseteq \mathbb{P}^{\binom{n}{k}-1}$. Choose a non-zero coordinate a_I and rescale so that $a_I = 1$. By definition $a_I = \det(M(V)_I) \neq 0$ hence this submatrix is invertible. Then the product $M(V)_I^{-1} M(V)$ results in a matrix representation of V that admits an identity matrix in the columns index by I. The remainder of the argument will show that there is a unique matrix representation mapping to a. Let M be a matrix reprsentation of some $V \in \mathbb{G}(k, n)$ such that $P_{k,n}(V) = a$ and such that M_I is the identity matrix. Let $j \in \{0, \ldots, n\} \setminus I$. For each $i \in I$, consider $I_i = (I \setminus \{i\}) \cup \{j\}$. In words we are replacing the i-th column of the identity matrix with the j-th column of M. From our map we have that

$$a_{I_i} = \det(M_{I_i})$$

for all $i \in I$. Observe that the determinant of this matrix is simply the i-th entry of M_j, the j-th column of M, up to a sign. Then this relation uniquely determines all entries of M_j. Since j was arbitrary, M is uniquely determined from a. That is using a we can reconstruct the entirety of M using the above relation. ℵ

Observe that using the homogeneous coordinates of the points we defined in the previous example with $I = \{1\}$ and $J = \{2, 3, 4\}$ gives us a polynomial

$$f_{I,J} = -x_{12}x_{34} + x_{13}x_{24} - x_{14}x_{23}$$

In fact this polynomial is unique to this particular choice of homogeneous components. We call these polynomials Plucker polynomials and we will use them to endow the Plucker map on the Grassmanians with the structure of a variety.

Proposition 2.14. *For any integer k and n with $0 \leq k \leq n$, we have*

$$P_{k,n}(\mathbb{G}(k,n)) = \mathcal{V}(f_{I,J} | I, J \subseteq \{1, \ldots, n\}, |I| = k-1, |J| = k+1)$$

The proof is ommited. Instead I'll, for the sake of time, finish of this section by saying that the Plucker map is an injection into this identity which allows us to think of subspaces in the grassmanian or as plucker coordinates because the changes in one correspond to changes in the other. There has to be a little bit of care about what you are doing in either department to ensure because modifying plucker coordinates is only modifying them as projective coordinates in the ambient space and it becomes necessary to ensure that you are still in the image of the Plucker map.

3 CH11 Quasiprojective Varieties

3.1 11.1 Quasiprojective Varieties

In this section we aim to further generalize the notion of varieties so that projective and affine varieites arrise as a special case of this notion. In order to do this we must being introducing some notions of topology. In particular we want to define the specific topology of $\mathbb{P}^n$ and characterize projective an affine varieties as the building blocks.

Definition 3.1 (Zariski Topology)**.** *The Zariski Topology on $\mathbb{P}^n$ is the topology whose closed sets, called zariski-closed, are projective varities.*

Of course as a direct consequence the open sets of this topology correspond to complements of projective varieties.

Definition 3.2 (Subspace Topology)**.** *Let $X \subseteq \mathbb{P}^n$ be a subset. The Zariski-topology on X is the subspace topology induced from the Zariski topology on $\mathbb{P}^n$. More explicitly, we say $Y \subseteq X$ is zariski-closed in X if $Y = Z \cap X$ for some projective variety $Z \subseteq \mathbb{P}^n$.*

Observe that if X is not a projective variety then $Y = X$ is Zariski-closed as $Y = \mathbb{P}^n \cap X$ and $\mathbb{P}^n$ is a projective variety although Y is not zariski-closed in X. Observe that if $X \subseteq \mathbb{P}^n$ is a projective variety then by definition subsets $Y \subseteq X$ are zariski-closed if there exists $Z \subseteq \mathbb{P}^n$ such that $Y = Z \cap X$. Becuase intersections of projective varieties are themselves projective varieties this implies that the Zariski closed subsets of X are are themselves projective varieties. The next definition is the generalization that we will work with for the remaining chapter.

Definition 3.3. *A subset $X \subseteq \mathbb{P}^n$ is called a quasiprojective variety if $X = Z \cap U$ where $Z \subseteq \mathbb{P}^n$ is Zarisk-closed and $U \subseteq \mathbb{P}^n$ is Zariski-open.*

Definition 3.4. *Let $X \subseteq \mathbb{P}^n$ be a projective variety. We say that $Y \subseteq X$ is a subvariety of X if $Y \subseteq \mathbb{P}^n$ is, itself, a quasiprojective variety.*

Proposition 3.1. *Let $X \subseteq \mathbb{P}^n$ be a projective variety. A subset $Y \subseteq X$ is a subvariety of X if and only if $Y = Z \cap U$ where Z is closed in X and U is open in X.*

Proof. Let $Y \subseteq X$ be a subvariety of X. Then by definition Y is a quasiprojective variety which implies there exists $Z, U \subseteq \mathbb{P}^n$ where Z is closed and U is open such that $Y = Z \cap U$. Since

$$Y = Y \cap X = (Z \cap X) \cap (U \cap X)$$

and $Z \cap X$ is intersection of closed sets hence closed and $U \cap X$ is intersection of open sets hence open in X. Now suppose $Y = Z \cap U$ where Z is closed and U is open in X. By definition this means that

$$Z = X \cap Z' \quad \text{and} \quad U = Z \cap U'$$

where Z' is closed in $\mathbb{P}^n$ and U' is open in $\mathbb{P}^n$. Since finite intersections of closed sets are closed and finite intersectinos of open sets are open sets, then

$$Y = Z \cap U = (Z' \cap X) \cap (U' \cap X)$$

is an intersection of a zariski closed and zariski open subset hence Y is a quasiprojective variety. ℵ

3.2 11.2 Closure and Irreducibility

There are a couple of usefull propositions though most of them are due to the topology rather than the geometry itself. I don't want to prove them as I had to continue reading and I will spend more time on the section on regular maps and functions on quasiprojective varieties.

3.3 11.3 Regular Maps and Regular Functions

With the generalization of varieities to quasiprojective varieties we aim to redevelop the notion of regular maps. Under this notion the special case of a regular map $F : X \to \mathbb{A}^1$ where $\mathbb{A}^1$ is identified with K, lets us develop the notion of regular functions on quasipojective varieties. Thankfully, the notion of regular functions on varieties lifts directly as a regular function on quasirpojective varieites. Categorically we can think of this notion as an embedding as {varieties} $\subset$ {quasirpojective varieties} and morphisms of one category are exactly an inclusion of morphisms into another. In the following definition we tackle the special case we mentioned above and identify $\mathbb{A}^1 \to K$, via a map $j_0(\mathbb{A}^1) \to K$ where $[a_0 : a_1] \mapsto \frac{a_1}{a_0}$.

Definition 3.5. *Let $X \subseteq \mathbb{P}^n$ be a quasiprojective variety. A function $F : X \to K$ is a regular function if, for every $p \in X$, there exist polynomiasl $f, g \in K[x_0, \ldots, x_n]$ homogeneous of the same degree such that $p \notin \mathcal{V}(g)$ and*

$$F(a) = \frac{f(a)}{g(a)} \qquad \forall a \in X \setminus \mathcal{V}(g)$$

Similar to a regular map, we have described functions that are locally described by rational functions. Observe that regular functions behave well with addition, multiplication, and scalar multiplication because at the local level these operations are reduced to operations on quotients of polynomials. In particular, given that a point doesn't vanish in one denominator nor the other paired with quotient operations ensures that by the end this point still doesn't vanish on the denominator polynomial.

Definition 3.6. *Let $K[X]$ denote the sets of all regular functions $F : X \to K$. This set naturally forms a K-algebra*

More precisely we can show that this ring is reduced, has no nilpotents. Let $X \subseteq \mathbb{P}^n$ be a quasiprojective variety. For $F \in K[X]$ and suppose that for some $n > 0$ that $F^n = 0$. The regular functio F is given by polynomials $f, g \in K[x_0, \ldots, x_n]$ such that

$$F(a) = \frac{f(a)}{g(a)} \qquad \forall a \in X \setminus \mathcal{V}(g).$$

Because we are in a K-algebra, the n-th product is a rational function of the form

$$F^n(a) = \frac{f^n(a)}{g^n(a)} = 0 \qquad \forall a \in X \setminus \mathcal{V}(g).$$

This implies that $f^n(a) = 0$, but we know that $K[x_0, \ldots, x_n]$ is a reduced K-algebra, hence $f(a) = 0$. This means that F must have been the zero map on $K[X]$. This local-global presentation means that at different points $p \in X$, we may end up with different quotients of polynomials describing F. While this isn't a problem that can be strictly resolved with closed, varieties, subsets of projective space by looking at open subsets we can provide a canonical quotient for the entire subset.

Proposition 3.2. *Let $X \subseteq \mathbb{P}^m$ be an open subset. Given a regular function $F : X \to K$, there exist polynomials $f, g \in K[x_0, \ldots, x_n]$ homogeneous of the same degree such that $X \cap \mathcal{V}(g) = \emptyset$ and*

$$F(a) = \frac{f(a)}{g(a)} \qquad \forall a \in X$$

While you can define many regular functions from X, this proposition ensures that each one has a canonical representation.

Proof. By definition there exist polynomials $f, g \in K[x_0, \ldots, x_m]$ such that for all $a \in X \setminus \mathcal{V}(g)$ that

$$F(a) = \frac{f(a)}{g(a)}$$

If f, g have shared factors h then we can express

$$F(a) = \frac{f_0(a)h(a)}{g_0(a)h(a)} \qquad \forall a \in X \setminus \mathcal{V}(g_0 h)$$

We may view F as the product of the two different regular functions in $K[X]$ where one is defined on the common factors of f and g, as a result is constant, and the other is defined on the factors that f and g do not have in common. This allows us to reduce regular functions on polynomials with common factors. Assume for the remainder of the proof that F is defined on f and g with common factors reduced. We aim to show that $X \cap \mathcal{V}(g) = \emptyset$ or equivalently that no irreducible factor of g vanishes on a point $p \in X$. Towards a contradiction suppose that g has an irreducible factor h which vanishes on $p \in X$. By definition, there exists polynomials $f_p, g_p \in K[x_0, \ldots, x_m]$ of the same homogeneous degree, where $p \notin \mathcal{V}(g_p)$, such that

$$F(a) = \frac{f_p(a)}{g_p(a)} \qquad \forall a \in X \setminus \mathcal{V}(g_p).$$

Observe that for $a \in X \setminus (\mathcal{V}(g) \cup \mathcal{V}(g_p)))$ that

$$\frac{f(a)}{g(a)} = \frac{f_p(a)}{g_p(a)}$$

which implies that $f_p g - g_p f \in \mathcal{I}(X \setminus (\mathcal{V}(g) \cup \mathcal{V}(g_p)))$. Obsere that $X \setminus (\mathcal{V}(g) \cup \mathcal{V}(g_p)))$ is a non-empty open subset of $\mathbb{P}^m$ implying it must be dense hence

$$\mathcal{I}(X \setminus (\mathcal{V}(g) \cup \mathcal{V}(g_p))) = \mathcal{I}(\mathbb{P}^m) = \{0\}$$

hence

$$f_p g - g_p f = 0.$$

By construction, $g_p(p) \neq 0$ and $g(p) = 0$ which implies

$$f_p(p)g(p) - g_p(p)f(p) = g_p(p)f(p) = 0$$

hence $f(p) = 0$. This means that when $h(p) = 0$ then $f(p) = 0$, i.e. $\mathcal{V}(h) \cap X \subseteq \mathcal{V}(f)$. Note that because $\mathcal{V}(h) \cap X$ is a non-empty open subset of $\mathcal{V}(h)$ then by taking closure $\mathcal{V}(h) \subseteq \mathcal{V}(f)$. This implies that $\mathcal{I}(\mathcal{V}(h)) \supseteq \mathcal{I}(\mathcal{V}(f))$ and by the Nullstelensatz,

$$\sqrt{\langle h \rangle} \supseteq \sqrt{f}.$$

Since h is irreducible, $\langle h \rangle$ is prime and $\sqrt{\langle h \rangle} = \langle h \rangle$, therefore $f = gh$ for some $g \in K[x_0, \ldots, x_m]$ which means that h must divide f. We have shown that f and g have a common factor h contradicting the assumption that F was in it's most reduced form. This means that the most reduced form of a regular function is the canonical global polynomial ratio. ℵ

Applying this proposition, consider a regular function $F : \mathbb{P}^n \to K$, the only polynomials which don't vanish on $\mathbb{P}^m$ must be constant forcing g to be a constant polynomial. Because f must be of the same homogeneous degree it is also constant. Then F must be a constant function. In fact this implies that $K[\mathbb{P}^m] = K$ as each function on the total projective space is constant. Note that in the beginning of the book we develop the ring of coordinate functions, in particular the ring of coordinate functions $K[\mathbb{A}^n] \cong K[x_0, \ldots, x_n]$.

Example Let $X = \mathbb{P}^1 \setminus \mathcal{V}(xy)$ and observe that by the above proposition, there exists an $f, g \in K[x,y]$ of the same homogeneous degree that such that for all $a \in X \setminus \mathcal{V}(xy)$, $g(a) \neq 0$. This forces g to be of the for $x^i y^j$ for some $i, j > 0$. Then by the previous proposition any regular function $F : X \to K$ in $K[X]$ is given by

$$F([a:b]) = \frac{f(a,b)}{a^i b^j} \qquad \forall [a:b] \in X \setminus \mathcal{V}(xy)$$

Then the denominator is exactly a polynomial $g(x,y) = x^i y^j$ and the numerator is some polynomial homogeneous in degree $i+j$. Define the following function $K[X] \to K[x, x^{-1}]$ where

$$F(x,y) \mapsto F(x,1).$$

Observe that for a regular function $F(x,y) = \frac{f(x,y)}{x^i y^j}$ the evaluation at $(x,1)$ gives a rational function

$$\frac{f(x,1)}{x^i}.$$

This ration can be seperate into a sum of polynomials $h(x) + p(x)$ where $h(x)$ consists of all rational terms where the degree of the numerator is greater than i and $p(x)$ contains all terms whose numerator has degree less than i. It is clear that $h(x) \in K[x]$ and $p(x) \in K[x^{-1}]$ and their sum must be in $K[x, x^{-1}]$. To define the inverse consider that any Laurent polynomial f is of the form

$$f = \sum_{i=1}^{k} a_i x^{\alpha_i} + \sum_{i=0}^{m} b_i x^{\beta_i}.$$

for $\alpha_1, \ldots, \alpha_k, \beta_1, \ldots, \beta_m \in \mathbb{N}$. Take $\gamma = \alpha_1 \cdots \alpha_k$ then f can be expressed as a polynomial

$$\frac{g(x)}{x^\gamma}.$$

We define the inverse $K[x, x^{-1}] \to K[X]$ that takes the common denomintor form of any polynomial

$$\frac{f(x)}{x^\gamma} \mapsto \frac{y^d f(x/y)}{y^{d-j} x^j}$$

where d is the maximal degree of $f(x)$. We can see immediately that this is a well defined rational function of homogeneous degree d in the numerator and denominator. Similarly it's clear that this is an inverse to the forward direction defined earlier. It remains to show that this is a K-algebra homomorphism. This property is easy enough to check because it falls out of the behavior of operations on fractions. Then we can conclude that $K[X] \cong K[x, x^{-1}|]$.

Similar to regular maps, an isomorphism of quasiprojective varieties is predicated only the existence of a regular function with an inverse between the two varieties. Naturally we are curious when an isomorphism of projective varieties is preserved in their rings of regular functions.

Definition 3.7. *Let X and Y be quasiprojective varieites, and let $F : X \to Y$ be a regular function. The pullback homomorphism induced by F is the function $F^* : K[Y] \to K[X]$ such that*

$$F^*(G) = G \circ F$$

Proposition 3.3. *If $X \cong Y$ then $K[X] \cong K[Y]$.*

Proof. *exercise* ℵ

Note that with affine varieties we had this statement as an if and only if. As $K[\mathbb{P}^m] \cong K \cong K[\mathbb{P}^n]$ yet $\mathbb{P}^m \ncong \mathbb{P}^n$.

3.4 11.4 Affine Varieties Revisted

We are building up to the culimnating proof that establishes that the ring of regular function is isomorphic to the coordinate ring of an affine variety. The first thing we do is establish that regular function on affine vareities are actually polynomial. This allows us to reinterpret what it means to be an affine variety in the quasiprojective setting.

Proposition 3.4. *Let $X_0 \subseteq \mathbb{A}^m$ be an affine variety and let $X = j_0(X_0) \subseteq \mathbb{P}^m$ be the associated quasiprojective variety. Then there is a canonical isomorphism $K[X] = K[X_0]$ between the ring of regular functions on X and the coordinate ring X_0 given by sending F to $F_0 = F \circ j_0 \in K[X_0]$.*

Proof. Let $F : X \to K$ be a regular function, the composition $F_0 = F \circ j_0$ means that for $a \in X$,

$$F_0(a) = F([1 : a_1 : \cdots : a_n])$$

It remains to prove that this map is polynomial. By definition of regular functions for any $p \in X_0$ there exist polynomials $f_p, g_p \in K[x_1, \ldots, x_n]$ such that

$$F_0(a) = \frac{f_p(a)}{g_p(a)} \quad \forall a \in X_0 \setminus \mathcal{V}(g_p)$$

Note that we may assume that f_p and g_p vansih on all of Z. Otherwise we can multiply f_p and g_p by a polynomial vanishing on all of Z but not at p. For each $p \in X_0$, choose a pair (f_p, g_p) for every $p \in X_0$. For all $g, q \in X_0$, observe that

$$\frac{f_p(a)}{g_p(a)} = F_0(a) = \frac{f_q(a)}{g_q(a)}$$

implying that $f_p g_q = f_q g_p$ if $p \notin Z$ or $q \notin Z$. Suppose that $p, q \in Z$, then this equation holds for for all $Z \setminus \mathcal{V}(g_p g_q)$, but this being a dense subset of Z this must be true for all Z. Now define the ideal

$$J = \langle g_p : p \in X_0 \rangle$$

Because every generator is a polynomial that doesn't vanish on some point of X_0, then

$$\mathcal{V}(J + \mathcal{I}_{\mathbb{A}}(X_0)) = \mathcal{V}(J) \cap X_0 = \emptyset.$$

By the Nullstellensatz, this implies that $J + \mathcal{I}_{\mathbb{A}}(X_0) = K[x_1, \ldots, x_n]$ and in particular that $1 \in J + \mathcal{I}_{\mathbb{A}}(X_0)$. Therefore

$$1 = h_1 g_{p_1} + \cdots h_k g_{p_k} + f$$

for $h_1, \ldots, h_k \in K[x_0, \ldots, x_n]$, $p_1, \ldots, p_k \in X_0$ and $f \in \mathcal{I}_{\mathbb{A}}(X_0)$. Then for any $a \in X_0$ at least one g_{p_i} is nonzero on a so we assume without loss of generality that $g_{p_1}(a) \neq 0$. Utilizing that for any p_i, that $f_{p_i} g_{p_1} = f_{p_1} g_{p_i}$, then

$$F_0(a) = \frac{f_{p_1}}{g_{p_1}} \tag{23}$$

$$= f(a) + \sum_{i=1}^{k} h_i(a) g_{p_i}(a) \frac{f_{p_1}}{g_{p_1}} \tag{24}$$

$$= \sum_{i=1}^{k} h_i(a) \frac{f_{p_i} g_{p_1}}{f_{p_1}} \frac{f_{p_1}}{g_{p_1}} \tag{25}$$

$$= \sum_{i=1}^{k} h_i(a) f_{p_i}(a) \tag{26}$$

which is clearly a polynomial map. To invert this K-algebra homomorphism we take a $F_0 \in K[X_0]$ and choose a representative such that $F_0 = [f_0]$ for $f_0 \in K[x_1, \ldots, x_n]$. Then we can homogenize it to obtain a homogeneous polynomial $f \in K[x_0, \ldots, x_n]$ of degree d. Taking $F = f/x_0^d$ gives us a regular function on $K[X]$ which can be dehomogenized by j_0 to get back to F_0. ℵ

With this proposition we have the machinery to show that regular maps on affine varieties are polynomial maps.

Proposition 3.5. *Let $X_0 \subseteq \mathbb{A}^m$ and $Y_0 \subseteq \mathbb{A}^n$ be affine varieties and let $X = j_0(X_0) \subseteq \mathbb{P}^m$ and $Y = j_0(Y_0) \subseteq \mathbb{P}^n$ be the corresponding projective varieties. There is a bijection*

$$\{\textit{Regular Maps } X \to Y\} \to \{\textit{Polynomial Maps } X_0 \to Y_0\}$$

given by sending F to $F_0 = j_0^{-1} \circ F \circ j_0$.

Proof. If $F : X \to Y$ is a regular map. Then by definition

$$(j_0^{-1} \circ F)(a) = (\tilde{F}_1(a), \ldots, \tilde{F}_n(a))$$

where $\tilde{F}_i : X \to K$ is a regular function on X. The previous theorem implies that $\tilde{F}_i \circ j_0 \in K[X_0]$ is a polynomial map for all i. Therefore

$$(j_0^{-1} \circ F \circ j_0)(a) = ((\tilde{F}_1 \circ j_0)(a), \ldots, (\tilde{F}_n \circ j_0)(a))$$

is a polynomial map $X_0 \to Y_0$. ℵ

From this point onwards we need to reframe how we think about affine varieties. We can identify affine varieties with quasiprojective varieties but this identification process is no longer invariant under isomorphism. To recover this invariance we broaden what we call an affine variety:

Definition 3.8. *A quasiprojective variety $X \subseteq \mathbb{P}^m$ is called an affine variety if X is isomorphic to a quasiprojective variety Y such that $Y = j_0(Y_0)$ for some affine variety $Y_0 \subseteq \mathbb{A}^n$.*

For example consider the quasiprojective variety $X = \mathbb{A}^1 \setminus \{0\}$. This is not an affine variety by it is isomorphic to

$$Y = \mathcal{V}(w^2 - xy) \setminus \mathcal{V}(w) \subseteq \mathbb{P}^2$$

which can be identified with $j_0(Y_0)$ where $Y_0 = \mathcal{V}(1 - xy) \subset \mathbb{A}^2$. So while X does not arise from an affine variety it is isomorphic to a quasiprojective variety that does. This motivates our desire to think about isomorphism classes of quasiprojective varieties that arise as affine varieties rather than only those that arise from affine varieties.

3.5 11.5 Affine Opens and Local Properties

The goal of this section is to show that quasiprojective varieities are locally affine, that is every open subset of a quasiprojective variety is isomorphic to an affine variety. The first step is to show that to study non-vanishing sets of closed subsets of affine space.

Proposition 3.6. *If $X \subseteq \mathbb{A}^n$ is a Zariski-closed set and $f \in K[x_1, \ldots, x_n]$, then $U(f) = X \setminus \mathcal{V}(f)$ is an affine variety*

Proof. *Have questions about book proof with the lifint of the affine map to a map between the projective closures of the varieties* ℵ

With this result it can be the following proposition can be established. This proposition is really stronger than what we want. We show that a quasirprojective variety can be covered by finitely many affine open subvarieties.

Proposition 3.7. *Let X be a quasiprojective variety. Then there exist finitely many affine open subvarieties $U_1, \ldots, U_k \subseteq X$ such that $X = U_1 \cup \cdots \cup U_k$. In particular for every $p \in X$, there exists an affine open subvariety $U \subseteq X$ where $p \in U$.*

Proof. Let X be a quasiprojective variety, then $X = Y \setminus Z$, where Y, Z are Zariski-closed subsets of $\mathbb{P}^n$. Then Z can be written $Z = \mathcal{V}(f_1, \ldots, f_k)$ where $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ are polynomials of the same homogeneous degree. Note that $Z \subseteq \mathcal{V}(f_i)$, hence

$$X \cap (\mathcal{V}(f_i))^c = X \cap Z^c \cap (V(f_i))^c = Y \setminus \mathcal{V}(f_i)$$

which implies that $Y \setminus \mathcal{V}(f_i)$ is an open subvariety of X. Observe that $U_{ij} = (Y \setminus \mathcal{V}(f_i)) \cap \mathbb{A}^n_j$ is an open subvariety of X and, by reordering, $U_{ij} = (Y \cap \mathbb{A}^n_j) \setminus \mathcal{V}(f_i)$. As both Y and $\mathbb{A}^n_j$ are closed then by application of the previous propsoition U_{ij} is an affine variety. It remains to show that

$$X = \bigcup_{\substack{0 \leq i \leq k \\ 0 \leq j \leq n}} U_{ij}.$$

For the easy direction, suppose that a is the union. Then there exists some i, j such that $a \in (Y \setminus \mathcal{V}(f_i)) \cap \mathbb{A}^n_j$and $f_i(a) \neq 0$. This implies that $a \notin Z$ while $a \in Y$ hence $a \in Y \setminus Z = X$. Now suppose that $a \in X$, then $a \notin Z$ so there exists an i such that $f_i(a) \neq 0$. Therefore $a \in Y \setminus \mathcal{V}(f_i)$, as a must have at least one non-zero component a_j, then $a \in (Y \setminus \mathcal{V}(f_i)) \cap \mathbb{A}^n_j$. ℵ

Remark 3.1. This proof is quite literally telling you how to consruct the affine open cover. The difficulty becomes more about identifying which affine spaces you are isomorphic to rather than constructing the cover itself.

We can now say that every point in a quasiprojective neighborhood has an affine neighborhood. Looking back at an old example $X = \mathbb{A}^2 \setminus \{0, 0\}$ which a quasiprojecive variety that is not affine. Let

$$U = X \setminus \mathcal{V}(x) = \mathbb{A}^2 \setminus \mathcal{V}(x)$$

and

$$V = X \setminus \mathcal{V}(y) = \mathbb{A}^2 \setminus \mathcal{V}(y)$$

are by the first propsoition of this chapter affine varieties. Since $X = U \cup V$, then any point in this quasiprojective variety is contained in at least one of these affine varieties making X locally affine.

Upon establishing the final result we can present the idea of a local property. That is if for all p in a local topological space X, there exists a set $x \in U \subseteq X$ with property P, then X has property P. Observe that the previous proposition establishes quasiprojective varieites as local topological spaces. Even more powefully, the open subset containing any point of a quasiprojective variety is isomorphic to an affine variety allowing us to port over some of the the theory established on affine varieites. This unlocks ways for us to study properties like dimension, smoothness (in the next chapter), closedness, and regularity which we will establish below.

Proposition 3.8. *Let Y be a topological space and $X \subseteq Y$ be a subset. If every $p \in X$ is contained in a set $U_p \subseteq Y$ such that $X \cap U_p$ is closed in Y then X is closed in Y.*

Proposition 3.9. *Let $X \subseteq \mathbb{P}^n$ and $Y \subseteq \mathbb{P}^m$ be quasprojective varieties and $F : X \to Y$ be a function. If every $p \in X$ is contained in an open set $U_p \subseteq X$ such that $F|_{U_p}$ is regular. Then F is regular.*

Proof. Let $F : X \to Y$ be a map of quasprojective varieites. Using the local nature of X, then for $p \in X$ there exist an affine open subvariety U_p where $p \in U_p$. Let $g_1, \ldots, g_k \in K[x_0, \ldots, x_n]$ be polynomials of the same homogeneous degree such that $p \notin \mathcal{V}_g = \mathcal{V}(g_1, \ldots, g_n)$. Then

$$F(a) = [g_0(a) : \cdots : g_n(a)] \quad \forall a \in U_p \setminus \mathcal{V}_g$$

is regular on U_p. The goal is to show that this regularity extends to all of X. Let f be a homogeneous polynomial in $K[x_0, \ldots, x_n]$ that vanishes at $X \setminus U_p$ and $f(p) \neq 0$. Let $f_i = fg_i$ and observe that

$$[f_0 : \cdots : f_n] = [g_0 : \cdots : g_n]$$

agree on their domains as the first simply a scalar multiple of the latter. As $\mathcal{V}_g \subseteq \mathcal{V}(f_1, \ldots, f_n)$ then the domain of the latter can be extended as $X \setminus \mathcal{V}_f = U_p \setminus \mathcal{V}_f$. As $p \in U_p$, then $p \notin X \setminus \mathcal{V}_f$, hence the map

$$F(a) = [f_0(a) : \cdots : f_n(a)] \quad \forall a \in X \setminus \mathcal{V}_f$$

is regular at p. As we did this for an arbitrary point it follows that F is regular on all of X. ℵ

Summary 3.1

In this section we further expound on the local behaviors of quasirpojective varieites. We find that a quasiprojective variety is locally affine, and even better a quasirpojective variety can be covered by finitely many affine-open subvarieties. This is essentially extending the quasicompactness of affine and projective varieties into the case of quasiprojective varieties. The local nature of quasiprojective varieites will allow us to discuss properties by looking at local affine open peices rather than at the global variety. In particular we can port over the intrinsic properties of affine varieties that we discussed in the beginning of the book because each local peice is isomorphic to affine varieites.

3.6 11.6 Continuity of Regular Maps

We have spent this time talking about projective space under this new Zariski Topology and used openness, closedness, and locality to characterize the global behavior of quasiprojective varieties. Now we come to the standard topological notion of continuity. We ask what functions are continuous on projective varieites and we see that the answer lies in the familiar regular functions. In fact, an isomorphism of varieities under a regular function with a regular inverse is exactly a continuous function with a continuous inverse.

Proposition 3.1

Let $F : X \to Y$ be a regular map of quasiprojective varieties. If Z is closed in Y, then $F^{-1}(Z)$ is closed in X.

Proposition 3.2

Let $F : X \to Y$ be an isomorphism of quasiprojective varieites and $Z \subseteq X$, the the following assertions hold:

1. Z is open in X if and only if $F(Z)$ is open in Y.
2. Z is closed in X if and only if $F(Z)$ is closed in Y.
3. Z is a subvariety of X if and only if $F(Z)$ is a subvariety in Y.

If Z is a subvariety in X, then the restriction $F|_Z : Z \to F(Z)$ is an isomorphism of quasiprojective varieties.

Proposition 3.3

Let $X \subseteq \mathbb{P}^n$ and $Y \subseteq \mathbb{P}^m$ be quasiprojective varieites. A map $F : X \to Y$ is regular if and only if the following conditions hold:

1. F is continuous with respect to the Zariski topology, and
2. For every open set $V \subseteq Y$ and for every regular function $G \in K[V]$, then $G \circ F$ is regular on $F^{-1}(V)$.

Proposition 3.4

Let $F : X \to Y$ be a map of quasiprojective varieties. F is an isomorphism if and only if the following conditions hold:

1. F is a homeomorphism with respect to the Zariski topology, and
2. For every open set $V \subseteq Y$, the map $G \mapsto G \circ F$ is a well defined K-algebra homomorphism

$K[V]$ to $K[F^{-1}(V)]$.

Summary: Characterization of regular maps as continuous. Essentially telling us that the set theoretic intuition we have built about the preservation of certain topological properties under continuous maps can be used.

3.7 11.7 Products and Graphs

Harkening back to the Segre map, we aim to present this map again and ensures that this map is well defined on a pair of quasiprojective varieites.

Proposition 3.5

Products of Quasiprojective Varieties If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^m$ are quasiprojective varieties, then the Segre map S_m, n restricts to an injection of $X \times Y$ as a quasirprojective variety in $\mathbb{P}^{(n+1)(m+1)-1}$.

Proof. We remind ourselves that the quasiprojective variety is an injection on projective varieties, closed subsets. As X and Y are quasiprojective let $X = Z \setminus Z'$ and $Y = W \setminus W'$ where $Z, Z' \subseteq \mathbb{P}^m$ and $W, W' \subseteq \mathbb{P}^n$ are closed in their respective ambient projective spaces. Then,

$$X \times Y = (Z \times W) \setminus ((Z \times W') \cup (Z' \times W))$$

and under the segre map

$$S_{m,n}(X \times Y) = S_{m,n}(Z \times W) \setminus (S_{m,n}(Z \times W') \cup S_{m,n}(Z' \times W)).$$

As each of the products in the right term consists of closed varieties, and closure is preserved under finite union, then the image of the Segre map is a quasiprojective variety. ℵ

With this expanded notion we can now make sense of applying the segre map to products off projective and affine varieties. For example $\mathbb{P}^1 \times \mathbb{A}^2$. Note that $\mathbb{A}^2 = \mathbb{P}^2 \setminus \mathcal{V}(y_0)$, then

$$\S_{1,2}(\mathbb{P}^1 \times \mathbb{A}^2) = S_{1,2}(\mathbb{P}^1 \times \mathbb{P}^2) = S_{1,2}(\mathbb{P}^1 \times (\mathbb{P}^2 \setminus \mathcal{V}(y_0))) \setminus (\mathbb{P}^1 \times \mathcal{V}(y_0))$$

Normally,

$$S_{1,2}([a_0 : a_1], [b_0 : b_1 : b_2]) = [a_0b_0 : a_0b_1 : a_0b_2 : a_1b_0 : a_1b_1 : a_1b_2]$$

then the removal of this closed subset above results in

$$S_{1,2}(\mathbb{P}^1 \times \mathbb{A}^2) = S_{1,2}(\mathbb{P}^1 \times \mathbb{P}^2) \setminus \mathcal{V}(z_{00}, z_{10}).$$

Remark 3.2. In genearl, a pair of closed quasiprojective varieties X, Y is sent to the closed set $S_{m,n}(X \times Y)$. Similarly, open sets are sent to open sets where $S_{m,n} \cap (\mathcal{V}(\mathcal{S}))^c$, as opposed to the regular case where the intersection is not with the complement but with the closed variety.

Definition 3.1

Let $F : X \to Y$ be a regular function. The graph of a regular function is the set

$$\Gamma_F = \{(a, b) \in X \times Y \mid b = F(a)\} \subseteq X \times Y$$

For example let $X = \mathbb{P}^1 \setminus \mathcal{V}(x_0x_1)$ and $Y = \mathbb{P}^1$. Then the map $F : X \to Y$ is given by

$$F([a_0 : a_1]) = [a_0a_1 :^2_0 + a^2_1]$$

has the graph

$$\Gamma_F = \{[a_0 : a_1 : b_0 : b_1] \in X \times Y \mid [b_0 : b_1] = [a_0a_1 : a_0^2 + a_1^2]\}.$$

The condition on the set is equivalent to the cross product condition $b_0(a_0^2 + a_1^2) = b_1a_0a_1$ then we can express the graph as a variety

$$\Gamma_F = (X \times Y) \cap \mathcal{V}(y_0(x_0^2 + x_1^2) - y_1x_0x_1) \subseteq \mathbb{P}^1 \times \mathbb{P}^2.$$

More generally it is possible to show that the graph of regular function is always a closed subset.

Proposition 3.6

Let $X \subseteq \mathbb{P}^n$ and $Y \subseteq \mathbb{P}^m$ be quasiprojective varieties and $F : X \to Y$ a regular function. Then Γ_F is closed in $X \times Y$.

Proof. By regularity for $p \in X$, there exists polynomials $f_0, \ldots, f_m \in K[x_0, \ldots, x_n]$ such that $p \notin \mathcal{V}(f_0, \ldots, f_m)$. Note that we may view $f_0, \ldots, f_m \in K[x_0, \ldots, x_m, y_0, \ldots, y_n]$ depending only on $x_0, \ldots, x_n$. Now define

$$U_p = (X \times Y) \setminus \mathcal{V}(f_0, \ldots, f_m)$$

which contains the point (p, q). Observe that $\Gamma_F \cap U_p$ contains pairs of projective coordinates (a, b) such that $a \notin \mathcal{V}(f_0, \ldots, f_n)$ and

$$[f_0(a) : \cdots : f_n(a)] = [b_0 : \cdots : b_n]$$

which is equivalent to the condition that $b_i f_j(a) - b_j f_i(a) = 0$. Now define

$$Z_p = \{y_i f_j - y_j f_j \mid 0 \leq i, j \leq n\} \subseteq \mathbb{P}^n \times \mathbb{P}^m$$

and we can see that if $(a, b) \in U_p \cap \Gamma_F$ then $(a, b) \in Z_p \cap U_p$. The reverse inclusion follows from exactly the same logic hence we can conclude that $\Gamma_F \cap U_p = Z_p \cap U_p$. Since Z_p is closed in U_p, then we have show that Γ_F is closed in U_p. Since closedness is local and we have shown this for an arbitary neighborhood this implies that Γ_F is closed in $X \times Y$. ℵ

The utility of this proposition is that it gives us a new characterization for the image of a regular function. Given a regular function $F : X \to Y$ of quasiprojective vareities, by applying the projection map on the second coordinate to Γ_F we can extract the image of a regular function.

Summary 3.2

The goal of this section is to introduce the generalization of graphs on continuous functions to graphs of regular functions. To do this it is necessary to expand our Segre map to apply to quasiprojective varieties. Thankfully that is not to difficult. Then showing that the graph is closed in general, paired with the projection allows us to create a new way to identify the image of a regular function.

3.8 11.8 Images of Projective Varieties

Proposition 3.7

If $X \subseteq \mathbb{P}^m$ closed and $Y \subseteq \mathbb{P}^n$ is quasiprojective, then the projection $\pi : X \times Y \to Y$ takes a closed set to a closed set in Y.

Rather than giving a proof (it can be found in Beggings in Algebraic Geometry pg. 353) I will instead attempt to provide a summary of the strategy in the proof.

Proof. Sketch: The beginning of this proof establishes that you can quickly generalize to show that you only need to consider the product $\mathbb{P}^m \times \mathbb{A}^n$ and this implies any other combination of closed and quasiprojective. The remainder of the proof will tackle this case.

1) Let $Z \subseteq \mathbb{P}^m \times \mathbb{A}^n$ be a closed subset. Then $Z = \mathcal{V}(f_1, \ldots, f_k)$ for polynomials $f_1, \ldots, f_k \in K[x_0, \ldots, x_m, y_1, \ldots, y_n]$. We characterize the pre-image of Z under the projection as the set

$$\pi^{-1}(Z) = \{(a, b) \in \mathbb{P}^m \times \mathbb{A}^n \mid f_i(a) = 0 \quad \forall i\}$$

2) Fixing a single point $b = (b_1, \ldots, b_n) \in \mathbb{A}^n$ we can characterize the fiber over particular points of $\mathbb{A}^n$. Define $f_i^b(x_0, \ldots, x_m) = f_i(x_0, \ldots, x_m, b_0, \ldots, b_m)$. Then fibers over b are empty when

$$\pi^{-1}(b) \cap Z = \emptyset \iff \mathcal{V}(f_0^b, \ldots, f_k^b) = \emptyset$$

3) Observe that the Projective Nullstellensatz implies that $I = \langle f_0^b, \ldots, f_k^b \rangle \supseteq \langle x_0, \ldots, x_m \rangle$. More accutely, there exists some $r \geq 0$ such that all monomials of degree r are contained in I. Define

$$M_r = \{\text{monomials of degree } r\}.$$

Observe that for $\alpha \in M_r$ corresponding to monomial x^α is contained in I and hence can be given by the sum

$$x^\alpha = \sum_{j=1}^{k} g_j f_j^b$$

where $g_j \in K[x_0, \ldots, x_m]$ for all j. This allows us to further recharacterize the fiber, i.e. $\pi^{-1}(b) \cap Z = \emptyset$ if and only if the above sum exists for all α.

4) Now we further refine the sum in order to a two parameter coefficient that will supply a matrix. Once we have a matrix condition we'll show that this conditions relies on the determinants giving us polynomial equations for when the fiber over a point is empty. Note that g_j can be expressed as the sum

$$g_j = \sum_{\beta} a_{\beta,j} x^\beta$$

then,

$$x^\alpha = \sum_{j=1}^{k} \sum_{\beta} a_{\beta,j} x^\beta f_j^b.$$

Since x^α is homogeneous, both terms in the summan must be homogeneous then take d_j to be the homogeneous degree of f_j^b and we see that $\beta \in M_{\beta - d_j}$. Examining individual monomials in the summand we see that

$$x^\beta f_j^b = \sum_{\alpha} c^b_{(\beta,j),\alpha} x^\alpha$$

and observe that the coordinates $c_{(\beta,j),\alpha}$ can be used to populate matrix C_r^b where the columns are indexed by M_r and the rows are indexed by pairs (β, j) with $\beta \in M_{r-d_j}$.

r) The final observation is to see that each term $x^\beta f_j^b$ is a linear combination of x^α where $\alpha \in M_r$. This implies that the standard basis vectors are in the row span C_r^b. In other words this means that $\text{rk}(C_r^b) = |M_r|$. Hence we can recharacterize the fiber over a point one more time

$$\pi^{-1}(b) \cap Z = \emptyset \iff \exists r \geq 0 \colon \text{rk}(C_b^r) = |M_r|$$

then negated

$$\pi^{-1}(b) \cap Z \neq \emptyset \iff \forall r \geq 0, \text{rk}(C_b^r) < |M_r|$$

which is precisely the conditions that all $|M_r| \times |M_r|$ minors of C_b^r vanish establishing a polynomial condition on the fiber over a point. Therefore $\pi(Z)$ is closed.

ℵ

This is the essential meat of the proof of the following proposition

Proposition 3.8

Let $F : X \to Y$ be a regular map of quasiprojective varieties, then $F(X)$ is closed in Y.

Proof. Let Γ_F be the graph of regular map F. We established that Γ_F is a closed subset of $X \times Y$, then the previous proposition ensures that $\pi(\Gamma_F)$ is closed in Y. Since $\pi(\Gamma_F) = F(X)$, then $F(X)$ is closed in Y. ℵ

From this proof follow two useful characterizations of closedness follow allowing us to determine whether something is or isn't a projective variety.

Proposition 3.9

If $X \subseteq \mathbb{P}^n$ and $Y \subseteq \mathbb{P}^m$ are isomorphic quasiprojective varieties. Then X is closed in $\mathbb{P}^n$ if and only if Y is closed in $\mathbb{P}^m$.

Proof. We established that $F(X)$ is closed in $\mathbb{P}^m$. Since this is an isomorphism and $F(X) = Y$ then Y must be closed in $\mathbb{P}^m$. The reverse follows via the exact same argument. ℵ

Extending this to the algebraic side, we saw previously that the ring of regular functions on $\mathbb{P}^m$, $K[\mathbb{P}^m]$, was exactly equivalent to K. We can refine this to show that the ring of regular functions on an irreducible projective variety are similarly uninteresting.

Proposition 3.10

Let X be an irreducible projective variety, then every regular function on X is constant. More succinctly $K[X] = K$.

Proof. Let $F : X \to K$ be a regular function, because we define regular functions as regular maps $F : X \to \mathbb{A}^1$ then $F(X)$ must be closed in $\mathbb{A}^1$. We have established long ago that the only closed subset of $\mathbb{A}^1$ are all of $\mathbb{A}^1$ of a finite number of points. If $F(X) = \mathbb{A}^1$, then the composition $(j_0 \circ F)(X) = \mathbb{P}^1$ is a map from $X \to \mathbb{P}^1$ where $F(X) = \mathbb{A}^1 \subset \mathbb{P}^1$ is not closed contradicting the closure of X. Then it follows that $F(X)$ must be a finite number of points in $\mathbb{A}^1$. Let $a_1, \ldots, a_\ell \in \mathbb{A}^1$ be the finite number of points in the image of F. Then

$$X = F^{-1}(a_1) \sqcup \cdots \sqcup F^{-1}(a_\ell)$$

contradicts the irreducibility of X. Hence the image of X under F must be a single point. ℵ

Summary 3.3

The goal of this section was to establish the proposition that projections of products of closed and quasirojective varieties are closed under the projection map to the first coordinate. The proof is involved and the stategy is to show that a more general case $\mathbb{P}^m \times \mathbb{A}^n$ will imply the statement. This allows us to establish that closedness is an intrinsic property of quasiprojective varieities. Note that we previously established that closed subsets of quasiprojecive varieties mapped to closed subsets. So we are now globalizing this statement for a pair of isomorphic quasiprojective varieties.

4 CH12 Culminating Topics

4.1 12.1 Rational Functions

The goal of the following section is to build up the notion of birational equivalence. If we consider an isomorphism of quasiprojective varieties, we notice that the isomorphism is rather strict in the supposition that this function must be defined over all of the domain. Birational functions weaken this condition by defining an "almost everywhere" presentation utilizing rational maps. What is fundamentally important is that birational equivalences preserve the rings of rational functions of varieties. We begin by presenting rational functions.

Definition 4.1: Rational Function on QP Variety

Let X be a quasiprojective variety. A rational function on X is an equivalence class of pairs (U, F), where $U \subseteq X$ is a dense open subset and F is a regular function on U, under the equivalence relation

$$(U, F) \sim (V, G) \iff F(a) = G(a) \quad \forall a \in U \cap V$$

I think the veracity of this equivalence relation is quite immediately checked thanks to the niceties of intersections. Similar to the presentation of regular functions, as rational function $F : X \dashrightarrow K$ where we identify K as the affine projection of $\mathbb{P}^1$. Naturally we can define the ring of rational functions:

Definition 4.2

Let X be a quasiprojective variety. The ring of rational functions on X, denoted $K(X)$, is the K-algebra of all rational functions $F : X \dashrightarrow K$.

Remark 4.1. The unit here is kind of strange. Define $F : X \to K$ to be a regular function such that $F(X) = 1$ then any dense open subset must now agree with this global $(X, 1)$ hence it is just the unit. Secondly the density requirement ensures that the intersection, which appears in the products of equivalence classes, is a non-empty open subset. If it were empty they we would be contradicting the density of the two chosen subsets.

Note that given an isomorphism of quasiprojective varieties $F : X \to Y$, then at the level of each pair (U, G), where U is a dense open subset, then the isomorphism of local K-algebras induced by the homomorphism raises to an isomorphism of rational rings of X and Y.

Example: In this example we want to justify that $K(\mathbb{P}^n) = K(x_1, \ldots, x_n)$. Our goal will be to show that given a dense open subset $U \subseteq \mathbb{P}^n$ equipped with a regular function F is $\sim$ to a pair with regular function given by a ratio of polynomials homogeous of the same degree.

Given any rational function on $\mathbb{P}^n$ with a regular representative $F : U \to K$ where U is a dense open subset of $\mathbb{P}^n$. As we saw previously, there exist polynomials $f, g \in K[x_0, \ldots, x_n]$ homogeneous of the same degree such that $F = f/g$ where $U \cap \mathcal{V}(g) = \emptyset$. Then,

$$(U, F) \sim (\mathbb{P}^n \setminus \mathcal{V}(g), f/g)$$

which we can see agrees on the interesection $U \cap (\mathbb{P}^n \setminus \mathcal{V}(g))$ establishing our first claim.

This allows us to identify $K(\mathbb{P}^n)$ with a subring of $K(x_0, \ldots, x_n)$ and upon homogenizing each polynomial at x_0 we get a canonical isomorphism $K(\mathbb{P}^n) = K(x_1, \ldots, x_n)$. Note that $\mathbb{A}^n$, a dense open subset of $\mathbb{P}^n$, has a ring of fractions identical to $K(x_1, \ldots, x_n)$.

Example 2: Let $X = \mathcal{V}(xy) \subseteq \mathbb{P}^2$ which is the union of two components $\mathcal{V}(x)$ and $\mathcal{V}(y)$. Define two rational functions $F : X \dashrightarrow K$ and $G : X \dashrightarrow K$ where

$$F([a : b : c]) = \frac{a}{c} \quad \text{and} \quad G([a : b : c]) = \frac{b}{c}.$$

Observe that the domain of F and G is $X \setminus \mathcal{V}(z)$ which is dense as we are removing a finite number of points from a projective variety. Observe that the domain of the product FG is also defined over $X \setminus \mathcal{V}(z)$, yet the ratio ab/c will always vanish as $ab = 0$ for all points in X. Therefore the ring of rational functions on X has zero divisors and is not a field.

What is particularly useful is that the ring of rational functions of a quasiprojective variety can be used as a detetctor of irreducibility.

Proposition 4.1

Let X be a quasiprojective variety. The ring of rational function $K(X)$ is a field if and only if X is an irreducible quasiprojective variety.

Before attempting a proof we notice the following if X is reducible. Given a dense open subset $U \subseteq X$ and regular function F representing an equivalence class $[(U, F)]$. If $K(X)$ were to be a field, then we should be able to take $U' = \{x \in U : F(x) \neq 0\}$ and define a new equivalence class $[(U', 1/F)]$. Yet as X is reducible, U' while non-empty and open is not guaranteed to be dense.

Looking back at our previous example, if we take $U = X \setminus \mathcal{V}(z)$ and set $U' = \{a \neq 0\} \cap U$, then we get

$$U' = \mathcal{V}(y) \cap U$$

which is not dense in X. Therefore the equivalence class required to determine the multiplicative inverse will not be in the ring of fractions.

Proof. ($\implies$) Assume X to be an irreducible quasirpojective variety. As $K(X)$ is already defined to be a commutative ring, in order to show that this is a field it remains to show that each equivalence class has a multiplicative inverse. Let $U \subseteq X$ be a dense open subset of X and F be a regular function on such that $[(U, F)] \neq 0$. Let $U' = \{x \in U : F(x) \neq 0\}$ and note that as $F(x) \neq 0$ for all X by assumption then U' is non-empty. It remains to show that U' is open in X to ensure that U' is dense.

As F is a regular function on an open subset U, then there exist polynomials f, g homogeneous of the same degree such that $F = f/g$ such that $U \cap \mathcal{V}(g) = \emptyset$. Then $x \in U$ such that $F(x) = f(x)/g(x) \neq 0$ are exactly points $x \notin \mathcal{V}(f)$ therefore

$$U' = U \setminus \mathcal{V}(f).$$

Since U' can be expressed as the intersection of two open subsets it must iself be open hence U' is a non-empty open subset of X and therefore dense. Then the rational function given by the representative $[(U', 1/F)]$ is well defined and the product

$$[(U, F)][(U', 1/F)] = [(U \cap U', F \cdot 1/F)] = [(U \cap U', 1)] = [(U', 1)].$$

Hence for any equivalence class we have a multiplicative inverse.

($\impliedby$) Assume that $K(X)$ is a field and suppose towards a contradiction that $X \subseteq \mathbb{P}^n$ is a reducible quasiprojective variety. Then there exists proper closed subsets Y, Z of X such that $X = Y \cup Z$. Consider a point $p \in Y \setminus Z$, and note that $p \notin Z$ implies that $p \notin X \cap \tilde{Z}$, where $\tilde{Z}$ is closed in $\mathbb{P}^n$. Therefore there exists a polynomial $f \in \mathcal{I}(\tilde{Z})$ such that f vanishes on all of Z except at p. *?* Let h be a polynomial of homogeneous degree equivalent to that of f such that h does not vanish uniformly on any irrducible component. *This is constructible, I have no clue as I have no clue apprently how to do anything.*. Then we have a regular function on $U = X \setminus \mathcal{V}(h)$ where

$$F(a) = \frac{f(a)}{h(a)} \quad \forall a \in U$$

which vanishes exactly on $Z \cap U$. Construct identically a regular function G such that G is defined on $U = X \setminus \mathcal{V}(h)$ and vanishes on $Y \cap U$. Then these are both rational functions over the same domain hence their product FG is well defined. Yet for $a \in X \setminus \mathcal{V}(h)$, if $a \in Y$ then $(FG)(a) = 0$ and if $a \in Z$ then $(FG)(a) = 0$ hence we have constructed two non-zero rational functions whose product is a zero function. Therefore $K(X)$ is not a field contradicting our assumption. ℵ

To show that this result makes any sense, we will show that the field of fractions of an irreducible Zariski-closed set is equivalent to the field of rational functions of the projective variety associated to it.

Proposition 4.2

Let $X_0 \subseteq \mathbb{A}^n$ be an irreducible closed set, and let $X = j_0(X_0) \subseteq \mathbb{P}^n$ be the corresponding quasiprojective variety. Then there is a canonical K-algebra isomorphism $K(X) = K(X_0)$.

Proof. Note that $K(X_0) = \mathrm{Frac}(K[X_0])$ then let F_0/G_0 be a rational function such that $F_0, G_0 \in K[X_0]$ and $G_0 \neq 0$. Let $F_0 = [f_0]$ and $G_0 = [g_0]$ with representatives $f_0, g_0 \in K[x_1, \ldots, x_n]$. Let $f, g \in K[x_0, \ldots, x_n]$ be homogeneous polynomials of the same degree that dehomogenize to f_0 and g_0 at x_0 respectively. Observe that

$$f/g : X \dashrightarrow K$$

is rational as the ratio f/g is regular on $X \setminus \mathcal{V}(g)$. Then our isomorphism $\phi : K(X_0) \to K(X)$ where $F_0/G_0 \mapsto f/g$ chosen as above.

Well-Defined: Let F_0', G_0' be an equivalent quotient such that $F_0'/G_0' = F_0/G_0$, then $F_0'G_0 = F_0G_0' \in K[X_0]$. Let $f_0', g_0' \in K[x_1, \ldots, x_n]$ be polynomials such that $F_0' = [f_0']$ and $G_0' = [g_0']$ then the previous equivalence implies that $f_0'g_0 - f_0g_0' \in \mathcal{I}(X_0)$. Then the rational functions f/g and f'/g' agree on

$$(X \setminus \mathcal{V}(g)) \cup (X \setminus \mathcal{V}(g')) = X \setminus \mathcal{V}(gg')$$

hence they are representatives of the same rational function. Therefore ϕ is well defined.

Injective: Suppose that $\phi(F_0/G_0) = f/g = 0 \in K(X)$, then f/g vanishes on a dense open set by definition. This implies that f vanishes on X which only happens if F_0 vanishes on X_0. Therefore $F_0/G_0 = 0 \in K(X_0)$.

Surjective: Let $F : X \dashrightarrow K$ be a rational function on X. For p in the domain of F by regularity of there exists polynomials $f, g \in K[x_0, \ldots, x_n]$ such that $p \notin \mathcal{V}(g)$ and $F = f/g$ on $X \setminus \mathcal{V}(g)$. By dehomogenizing f and g we get $f_0, x_0 \in K[x_1, \ldots, x_n]$ such that $\phi([f_0]/[g_0]) = F$.

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Proposition 4.3

Let X be a quasiprojective variety, let $U \subseteq X$ be a dense open subvariety of X. Then $K(X) = K(U)$.

Proof. Let U be a dense open subset, then a rational functino $F : U \dashrightarrow K$ defined on $V \subseteq U$ gives to a rational function $F : X \dashrightarrow K$ defined on $V \subseteq X$. Then this gives inclusion $K(U) \subseteq K(X)$. Let $G : X \dashrightarrow K$ be a rational function, then for a dense subset $W \subseteq X$ the intersection $W \cap U$ is dense and open in X. Hence

$$(U, G) \sim (U \cap W, G|_{U \cap W}).$$

Since $U \cap W$ is dense in U then G can be viewed as a rational function $U \dashrightarrow K$ hence $G \in K(U)$. ℵ

With this we can define a rational map between quasiprojective varieties which will act as a slight generalization of a regular map.

Definition 4.3

Let X and Y be quasiprojective varieties. A rational map $X \dashrightarrow Y$ is an equivalence class of pairs (U, F) where U is a dense open subset of X and $F : U \to Y$ is a regular map under the equivalence relation

$$(U, F) \sim (V, G) \iff F(a) = G(a) \quad \forall a \in U \cap V$$

With rational maps we can define our weak form of isomorphism. Quasiprojective varieties X and Y are considered birationally equivalent if there exist rational maps $F : X \dashrightarrow Y$ and $G : Y \dashrightarrow X$ such that $G \circ F = \mathrm{id}_X$ and $F \circ G = \mathrm{id}_Y$. Finally we arrive at the main result of birational equivalence.

Proposition 4.4

Let X and Y be quasiprojective varieties. If X and Y are birationally equivalent that $K(X)$ and $K(Y)$ are isomorphic as K-algebras.

Summary 4.1

In this chaper we define a weaker form of equivalence called birational equivalence. A birational function is now an equivalence class of regular functions which agree on intersections of dense open sets. Like regular fucntions we can gather them into a ring. This ring is a field when the quasiprojective variety we are dealing with is irreducible. Finally we show that birational equivalence induces an isomorphism on the rings of rational functions.

4.2 12.2 Dimension of Quasiprojective Varieties

The goal of this section is to define the dimension of a quasiprojective variety in a way that doesn't require us to entirely redevelope the theory from the affine setting. As we saw in 12.1, the ring of rational functions of a quasiprojective variety is conanically isomorphic to the fraction field of a dense open subset. Because the affine restrictions of a quasirprojective variety is dense in the quasiprojective variety, then it follows that we can apply dimension to the affine pieces of projective varieties.

Definition 4.4

Let X be an irreducible quasiprojective variety. The dimension of X is defined as

$$\dim(X) = \text{trdeg}_K(K(X)).$$

The dimension of a non-empty, reducible variety is the maximum of the dimension of each of its reducible components.

The first proposition we will establish is arguably the most important.

Proposition 4.5

Let X be a quasiprojective variety and $U \subseteq X$ be a dense open subset, then

$$\dim(U) = \dim(X).$$

This proposition tells us that it will make sense to study the affine pieces of a quasiprojective variety rather than redeveloping tools. This will then follow from the observation of the previous chapter which established that the fraction ring of a quasiprojective variety is equivalent to the fraction ring of a dense open subset of that variety.

Proof. We begin by assuming that X is an irreducible quasiprojective variety. Let $U \subseteq X$ be a dense open subset and we have established that $K(U) = K(X)$. Then, $\text{trdeg}_K(K(X)) = \text{trdeg}_K(K(U))$. Now suppose that X is reducible and $X_1, \ldots, X_r$ are the irreducible, closed subvarieties of X. Let U be a dense open subset of X, then

$$U = (X_1 \cap U) \cup \cdots \cup (X_r \cap U).$$

Since $X_i \cap U$ is a non-empty, open subset of X_i, then it is dense and irreducible in X_i hence $K(X_i \cap U) = K(X_i)$. Therefore the maximum dimension of an irreducble component of U must correspond to the maximum dimension of an irreducible component of X. ℵ

Using this we will compute the dimension of some examples.

Example: As the affine resitrciton of $\mathbb{P}^n$ is $\mathbb{A}^n$ and on top of this $\mathbb{A}^n$ is dense in $\mathbb{P}^n$ hence

$$\dim(\mathbb{P}^n) = \dim(\mathbb{A}^n) = n.$$

It is somewhat interesting that we are adding some extra information moving from $\mathbb{A}^n$ to $\mathbb{P}^n$ but not enough that we add another dimension. More so we are enriching the dimension we are in by including points at infinity rather than adding extra free parameters.

Example: Let $f \in K[x_0, \ldots, x_n]$ be a homogeneous polynomial. Then to compute $X = \mathcal{V}_{\mathbb{P}}(f) \subseteq \mathbb{P}^n$ observe that

$$X \cap \mathbb{A}_0^n = \mathcal{V}_{\mathbb{A}}(f(1, x_1, \ldots, x_n)) \subseteq \mathbb{A}^n$$

Since we can view f as a polynomial $f_0 \in K[x_1, \ldots, x_n]$ we can utilize affine dimension tools to establish that $\dim(\mathcal{V}_{\mathbb{A}}(f_0) = n - 1$ so it follows $\dim(X) = n - 1$. The follows form the Fundamental Theorem of Dimension Theory.

Now we aim to establish a generalization of the Fundamental Theorem of Dimension Theory in the context of quasiprojective varieties. To do this we start by presenting a new way to use the vanishing operator.

Definition 4.5: Local Hypersurface

Let X be a quasiprojective variety. We say that a non-empty subset $Y \subseteq X$ is a local hypersurface in X if for every $p \in X$ is contained in an open set $U \subseteq X$ such that

$$Y \cap U = \mathcal{V}(F) = \{a \in U \mid F(a) = 0\}$$

for a regular function $F \in K[U]$ that does not vanish on every irreducible component of U.

I am curious about why we are interested in this presentation of the vanishing operator. It will later show up in a proposition about the dimension of a local hypersurface. Will this have something to do with the latter portion of this book which will deal with lines on hypersurfaces. Note that the set $\mathcal{V}(F)$ can be viewed as the preimage of $\{0\} \in K$ of the regular function $F : U \to \mathbb{A}^1$. As this is the preimage of a closed set, the $\mathcal{V}(F) \subseteq U$ is closed in U by the continuity of regular maps. With local surfaces we can now present the generaliation of the Fundamental Theorem of Dimension Theory.

Proposition 4.6

Let X be a quasiprojective variety whose irreducible components all have the same dimension and let $Y \subseteq X$ be a local hypersurface of X. Then every irreducible component of Y has $\dim(X) - 1$.

Proof. Let Z be an irreducible component of Y. Then for $p \in Z$, there exists an open set $U \subseteq X$ such that $p \in U$ and regular function $F \in K[U]$ that does not vanish on any irreduicble component of U such that

$$Z \cap Y = \mathcal{V}(F).$$

As closed sets can be covered by affine opens, we can restrict further and assume that U is affine. Since $Z \cap U$ is irreducible then it must be contained in some irreducible component of U, calling this component V. That is V is an irreducible affine variety and furthermore $Z \cap V$ is an irreducible component of $Y \cap V = \mathcal{V}(F|_V)$. As $p \in \mathcal{V}(F|_V)$ and $F|_V \neq 0$, then this is a non-empty and not all of V. Then using the strong fundmantal theorem of dimensino theory, then

$$\dim(Z \cap V) = \dim(V) - 1.$$

Since V is an irreducible affine variety of X, then it must be dense in some component of X, all of which have the same dimension, then $\dim(V) = \dim(X)$. Since Z is irreducible then $\dim(Z) = \dim(Z \cap V)$. We conclude that

$$\dim(Z) = \dim(Z \cap V) = \dim(V) - 1 = \dim(X) - 1.$$

ℵ

In the Strong Fundamental Theorem of Dimension Theory, we established that cutting a hypersurface out of an irreducible affine variety reduced the dimension by exactly one. Similarly, we generalize to a local hypersurface to say that irreducible components of these hypersurfaces have their dimensino cut down by exactly one. The language of local hypersurfaces gives us a way to import this natural notion of dimension from affine land.

Proposition 4.7

Let $X \subseteq \mathbb{P}^n$ be a quasiprojective variety, all of whose irreducible components have the same dimension, and let $Y = \mathcal{V}(f_1, \ldots, f_k, F_1, \ldots, F_\ell) \subseteq X$ where $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ and $F_1, \ldots, F_\ell \in K[X]$. If $Y \neq \emptyset$, then

$$\dim(Y) \geq \dim(X) - k - \ell.$$

Proof. *exercise* ℵ

Finally, we can go back to the land of projective varieties and give a lower bound on their dimension.

Proposition 4.8

If $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ are non-constant homogeneous polynomials with $k \leq n$, then $\mathcal{V}_{\mathbb{P}}(f_1, \ldots, f_k) \neq \emptyset$ and

$$\dim(\mathcal{V}_{\mathbb{P}}(f_1, \ldots, f_k)) \geq n - k.$$

Nicely this definition of dimension aligns with the same lower bound established in the case of affine varieties.

Proof. *exercise* ℵ

Summary 4.2

content...